MIDTERMS: CMPA411 (COMPETENCY APPRAISAL 1) - MODULE #6: NURSING CARE MANAGEMENT FOR MATERNAL AND CHILD HEALTH

DEFINITION OF TERMS

— **OBSTETRICS** primarily concerned with the phenomenon and the management of pregnancy, labor and puerperium both under normal and abnormal circumstances. It is also derived from the Latin word "Obstetrix" which means midwife.

— **MATERNITY CARE** – is the care of pregnant women, her safe delivery, post natal examination, the care of the newly born infant and the maintenance of location.

— **MATERNITY NURSING** – involves direct personal care to maternity patients and their newborn infants or to related activities on their behalf during the various phases of the childbearing experience.

OBJECTIVE OF MATERNITY CARE:

To ensure that every expectant nursing mother maintains good health, learns childcare, has a normal delivery and bears healthy children.

ANATOMY AND PHYSIOLOGY OF FEMALE REPRODUCTIVE SYSTEM

FEMALE REPRODUCTIVE SYSTEM

— consist of the ovaries, the uterine tubes (or fallopian tubes)

— uterus, the vagina, the external genitalia, and the mammary glands

— UTERUS AND THE VAGINA are in the midline, with an ovary to each side of the uterus.

A. OVARIES

— **Broad ligament** – spreads out on both sides of the uterus and attaches to the ovaries and uterine tubes

— **Ovary** - is a small organ suspended in the pelvic cavity by two ligaments

— **Suspensory ligament** – extends from each ovary to the lateral body wall.

— **Ovarian ligament** – attaches the ovary to the superior margin of the uterus

— **Mesovarium** – folds of peritoneum.

OOGENESIS AND FERTILIZATION

➢ **OOGONIA**

— The cells form which oocytes develop

- **Primary oocyte** – oogonia that has stopped in prophase I
- **Secondary oocyte** – released when the first meiotic division is complete

— **Ovulation** – release of an oocyte from an ovary

— **Fertilization** – when a sperm cell penetrates the cytoplasm of a secondary oocyte

— **Zygote** – 23 chromosomes from the sperm + 23 chromosomes from the female gamet.

FOLLICLE DEVELOPMENT

— **Primordial follicle** – primary oocyte surrounded by granulosa cells (single layer of flat cells)

— **Primary follicles** – oocyte enlarges and the single layer of granulosa cells become enlarged and cuboidal

— **Zona pellucida** – a layer of clear material that is deposited around the primary oocyte

— **Secondary follicle** – flui-filled vesicles appear and a theca forms around the follicle.

— **Fluid filled spaces** - vesicles

— **Theca** – a capsule that forms around the follicle

— **Antrum** – a single, fluid-filled chamber

— **Cumulus cells** – mass of granulosa cells

— **Graafian follicle** – mature follicle

— **Corpus luteum** – remaining cells of the ruptured follicle are transformed into a glandular structure

— **Human chorionic gonadotropin hormone (hCG)** – the corpus luteum enlarges in response to this hormone.

B. UTERINE TUBES

1. Fallopian tube or oviduct

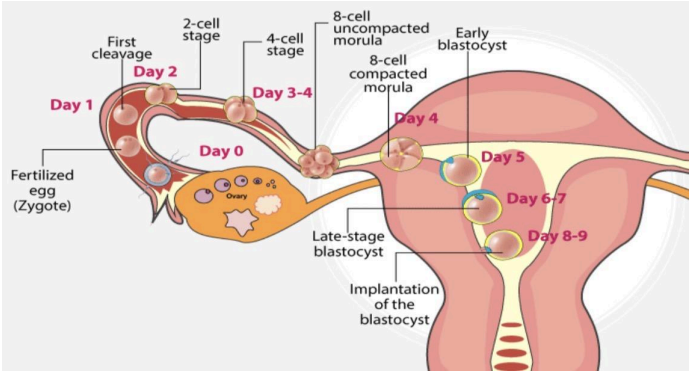
2. Receive the secondary oocyte

— **Fimbriae** – long, thin processes that surrounds each the opening of each uterine tube

— **Cilia on fimbriae** – sweep the oocyte into the uterine tube

— **Ampullae** – where fertilization usually occurs

— **Implantation** – process wherein the fertilized oocyte embeds itself in the uterine wall.



C. UTERUS

— **Fundus** – superior to the entrance of the uterine tubes

— **Body** – main part of the uterus

— **Cervix** – inferiorly, narrower part.

Layers of Uterus:

— **Perimetrium** – outer layer; serous layer of the uterus formed from visceral peritoneum.

— **Myometrium** – middle layer; muscular layer that accounts for the bulk of the uterine wall.

— **Endometrium** – innermost layer; consists of simple columnar epithelial cells with an underlying CT layer.

MENSTRUAL CYCLE

— The menstrual cycle is a monthly series of controlled hormonal changes that prepares a person's body for a potential pregnancy.

FOUR CORE PHASES

1. MENSTRUATION PHASE (DAYS 1-5)

— *What happens:* The thick uterine lining (endometrium) sheds if pregnancy did not occur in the previous cycle.
— *Hormone status:* Estrogen and progesterone drop to their lowest levels.
— *Symptoms:* Cramping, low energy, and bleeding lasting 3 to 7 days.

2. FOLLICULAR PHASE (DAYS 1–13)

— *What happens:* The pituitary gland secretes Follicle-Stimulating Hormone (FSH). This prompts multiple ovarian follicles to develop. Eventually, one dominant follicle matures an egg.
— *Hormone status:* Estrogen steadily rises, thickening the uterine lining to prepare for an embryo.

3. OVULATION PHASE (AROUND DAY 14)

— *What happens:* A sudden surge of Luteinizing Hormone (LH) triggers the dominant follicle to release its mature egg into the fallopian tube.
— *Fertile window:* This is the peak fertile period. The egg lives for roughly 24 hours

4. LUTEAL PHASE (DAYS 15–28)

— *What happens:* The ruptured follicle transforms into the corpus luteum. If fertilization does not happen, the corpus luteum breaks down.
— *Hormone status:* Progesterone peaks to maintain the womb's lining. If pregnancy fails, progesterone crashes, triggering the next period.

FERTILIZATION AND IMPLANTATION (The Beginning of Pregnancy)

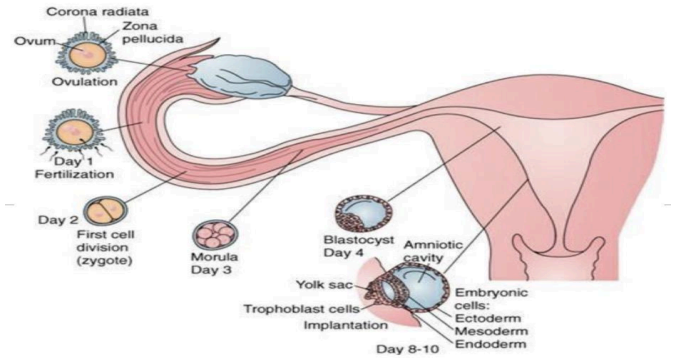
FERTILIZATION

— Fertilization occurs in the ampulla of the fallopian (uterine) tube when sperm and ovum unite.
— An ovum can be fertilized up to 72 hours after its release.

IMPLANTATION

— Fertilized egg becomes a zygote. The zygote is propelled toward the uterus and implants 6 to 10 days after ovulation.
— The blastocyst secretes chorionic gonadotropin to ensure that the corpus luteum remains viable and secretes estrogen and progesterone for the first 2 to 3 months of gestation.

STAGES OF FETAL DEVELOPMENT



— **Ovum** - from ovulation to fertilization
— **Zygote** - from fertilization to implantation
— **Embryo** - from implantation to 5-8 weeks
— **Fetus** - from 5-8 weeks until term
— **Conceptus** - developing embryo and placental structures throughout pregnancy.
— **Age of viability** - the earliest stage at which fetuses survive if they are born is generally accepted as 24 weeks or at the point a fetus weighs more than 500-600g

FETAL ENVIRONMENT

➢ AMNION

— Encloses the amniotic cavity
— Forms a fluid-filled sac that surrounds the embryo and later the fetus

➢ CHORION

— Is the outer membrane enclosing the amniotic cavity
— Becomes vascularized and forms the fetal part of the placenta

➢ AMNIOTIC FLUID

— Consists of 800 to 1200 mL by the end of pregnancy
— Surrounds, cushions, and protects the fetus and allows for fetal movement
— Maintains the body temperature of the fetus
— Contains fetal urine and is a measure of fetal kidney function
— The fetus modifies the amniotic fluid through the processes of swallowing, urinating, and movement of fluid through the respiratory tract.

➢ PLACENTA

— is a Latin term for “pancake” for its appearance came from the trophoblast tissues
— The placenta provides for exchange of nutrients and waste products between the fetus and birthing parent.
— The placenta begins to form at implantation; the structure is complete by week 12.
— It produces hormones to maintain pregnancy and assumes full responsibility for the production of these hormones by the 12th week of gestation.

FETAL CIRCULATION

A. UMBILICAL CORD

— It contains two arteries and one vein.
— The arteries carry deoxygenated blood and waste products from the fetus.
— The vein carries oxygenated blood and provides oxygen and nutrients to the fetus.

B. FETAL HEART RATE (FHR)

— FHR depends on gestational age; FHR is 160 to 170 beats per minute in the first trimester but slows with fetal growth to 110 to 160 beats per minute.
— FHR is about twice the maternal heart rate.

FETAL CIRCULATION BYPASS

THE THREE FETAL SHUNTS

➢ **Ductus Venosus**

— Connects the umbilical vein to the inferior vena cava (the main vein returning blood to the heart).
— Bypasses the liver.
— Sends oxygen-rich blood straight to the heart.

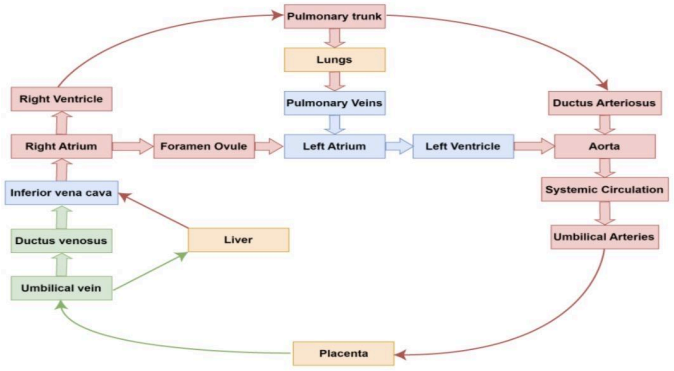
➢ **Foramen Ovale**

— A small hole between the right atrium and the left atrium of the heart.
— Bypasses the lungs.
— Pushes oxygenated blood directly into the left side of the heart to supply the brain and heart muscle

➢ **Ductus Arteriosus**

— Connects the pulmonary artery (leading to the lungs) directly to the aorta (leading to the body).
— Bypasses the non-working lungs.
— Protects the lungs from high-pressure blood flow.

FETAL CIRCULATION



PERINATAL PERIOD

➢ **GESTATION**

— Time from fertilization of the ovum until the date of delivery
— About 280 days
— Naegele's rule for estimating the date of delivery, also known as date of birth

➢ **GRAVIDITY AND PARITY**

— Clinical shorthand used by healthcare providers to record a patient's complete pregnancy history
— **Gravidity (G)**: The total number of times a person has been pregnant, regardless of the duration or outcome of the pregnancy.
— **Parity (P)**: The number of pregnancies that have reached viability

GTPAL
G - ravidity
T - erm births
P - arity
A - bortions or miscarriage
L - current Living children

SIGNS OF PREGNANCY

Presumptive	Probable	Positive
—Amenorrhea, — Nausea and vomiting — Fatigue — Breast tenderness — Quickening — Linea nigra — Melasma — Striae gravidarum	— Positive urine/blood pregnancy test — Chadwick's sign (bluish cervix), — Goodell's sign (soft cervix), — Hegar's sign (soft uterine isthmus). — Periodic uterine tightening and contractions. — Gestational sac — Fetal outline	— Hearing the fetal heartbeat (via Doppler) — Visualizing the fetus on an ultrasound — A clinician palpating actual fetal movement

PHYSIOLOGIC CHANGES THAT OCCUR DURING PREGNANCY

— **LOCAL** - confined to the reproductive organs
— **SYSTEMIC** - affecting the entire body.
— Both symptoms (subjective findings) and signs (objective findings) of the physiologic changes of pregnancy are used to diagnose and mark the progress of pregnancy.

A. REPRODUCTIVE SYSTEM CHANGES

— **Amenorrhea** - suppression of follicle-stimulating hormone (FSH) by rising estrogen levels.
— Cervix undergoes a more vascular and edematous appearance owing to the increased level of estrogen.
— Uterus increases in length, depth, width, weight, wall thickness, and volume.

- Length increases from approximately 6.5 to 32 cm.
- Depth increases from 2.5 to 22 cm.
- Width expands from 4 to 24 cm.
- Weight increases from 50 to 1000 g.

— ***UTERINE Changes:***

- End of the 12th week - above the symphysis pubis.
- 20th or 22nd week of pregnancy - it should reach the level of the umbilicus.
- 36th week - should touch the xiphoid process and can make breathing difficult.

— **Cervical changes** – becomes more vascular and edematous.

- Goodell's sign
- Chadwick's sign

— **Vaginal Changes** – increase vascularity of the vagina
— **Ovarian Changes** – active production of estrogen and progesterone
— ***BREAST Changes:***

- Feeling of fullness, tingling or tenderness because of increased estrogen level
- Breast size increase because of the growth in mammary alveoli and in fat deposit

- Areola of the NIPPLE DARKENS and the diameter increases from about 3.5 cm (1.5 inches) to 5cm or 7 cm (2 or 3 inches)
- By the 16th week, colostrum, the thin, watery, milk
- Increase vascularity
- Enlarge and protuberant nipples high –protein fluid that is the precursor of breast

B. INTEGUMENTARY SYSTEM CHANGES

— **STRIAE GRAVIDARUM** - pink or reddish streaks appearing on the sides of the abdominal wall
— **LINEA NIGRA** - narrow, brown line, running from the umbilicus to the symphysis pubis
- Darkened areas may appear on the face as well, particularly on the cheeks and across the nose.

— **MELASMA (CHLOASMA)** - the "mask of pregnancy."
— **VASCULAR SPIDERS OR TELANGIECTASES** - small, fiery-red branching spots; on the skin (thighs) - increased level of estrogen in the body.

C. ENDOCRINE CHANGES

— **FSH and LH** - decreases causing anovulation.
— **PROLACTIN** – prepare the breasts for lactation, increases in production of milk.
— **Melanocyte** – stimulating Hormones causes increase in skin pigment.
— **Human growth hormone** - increase to aid the fetus in growing.
— **Estrogen and Progesterone** - aids in uterine and breast enlargement.
— **Human Placental Lactogen** - increases glucose levels to supplement the growing fetus.
— **Relaxin** - increases to soften the cervix and collagen of joints.
— **THYROID AND PARATHYROID GLAND** - slight enlargement; increases the basal metabolic rate of a pregnant woman and for better consumption of calcium and vitamin D.
- Thyroid hormone production increases.

— **INSULIN** produced from the pancreas decreases EARLY in the pregnancy, thereby increasing glucose available for the fetus.
- Increase in insulin occurs in the first trimester because estrogen, progesterone and HPL have insulin antagonistic properties.

D. RESPIRATORY SYSTEM CHANGES

— Shortness of breath is also a common discomfort of pregnancy as the pregnant uterus pushes the diaphragm upward.
— The total oxygen consumption of a pregnant woman increases by 20%.
— Congestion, or "stuffiness," of the nasopharynx - response to increased estrogen levels.

E. CARDIOVASCULAR SYSTEM CHANGES

— Changes in the circulatory system are extremely significant to the health of the fetus - adequate placental and fetal circulation
— 30-50% increase in the total cardiac volume
— Physiologic Anemia of pregnancy may occur
— Increases heart rate
— Palpitations is common
— Edema and varicosities of the lower extremities

F. GASTROINTESTINAL SYSTEM CHANGES

— Slow emptying time of the stomach
— **Nausea and vomiting** - high levels of estrogen and human chorionic gonadotropin
— **Hyperemesis gravidarum** - severe nausea and excessive vomiting during pregnancy (abnormal)
— Decreased pH of the saliva
— **Constipation** - high level of progesterone during pregnancy slows the automatic waves of muscular contractions in the intestine
— **Hemorrhoids** - common due to constipation, pressure of the uterus, slow peristalsis

G. URINARY SYSTEM CHANGES

— **Glomerular Filtration** rate increases
- 60 or higher is in the normal range

— BUN and Creatinine decreases
— Renal threshold for sugar decreases - **renal glycosuria**
— Frequent urination in 1st trimester, normalizes in 2nd trimester, frequent urination in 3rd trimester

H. MUSCULOSKELETAL SYSTEM CHANGES

— Calcium and phosphorus needs are increased during pregnancy, because the fetal skeleton must be built.
— Gradual softening of a woman's pelvic ligaments and joints to facilitate passage of the baby through the pelvis at birth.
— Symphysis pubis widens for 3 to 4 mm.
— **'Pride of pregnancy'** or commonly **'lordosis'** occurs
- backache

— Muscle cramps are caused by decreased serum calcium levels

I. IMMUNE SYSTEM CHANGES

— Immunologic competency during pregnancy apparently decreases - to prevent a woman's body from rejecting the fetus
— **Immunoglobulin G (IgG)** production decreased, which can make a woman more prone to infection
— Increase in the white blood cell count may help to counteract the decrease in IgG response.

COMMON DISCOMFORTS OF PREGNANCY

➢ **Morning Sickness (Nausea & Vomiting):**

— Driven by surging pregnancy hormones
Relief:
— Eat small, frequent meals every 1 to 2 hours to avoid an empty stomach.
— Keep dry crackers or toast by your bed to eat before getting up.

— Ginger, vitamin B6, or peppermint tea can also soothe the stomach.

➢ **Heartburn & Indigestion:**

— Progesterone relaxes the valve between your stomach and esophagus, allowing gastric acid to creep up.

Relief:

— Avoid greasy, spicy, or trigger foods.
— Remain upright for at least an hour after eating and elevate head with extra pillows while sleeping

➢ **Constipation & Hemorrhoids:**

— Slower digestion and pressure from the growing uterus can lead to infrequent bowel movements and swollen veins.

Relief:

— Increase fluid intake, eat a high-fiber diet, and stay active.

➢ **Frequent Urination:**

— The growing uterus exerts direct pressure on your bladder, usually occurs on the first and third trimester

Relief:

— Limit fluids two hours before bedtime to minimize middle-of-the-night wakeups.
— Performing kegels exercise
— Voiding at regular intervals
— Sleeping side-lying at night
— Wearing perineal pads, if necessary

➢ **Breast Tenderness:**

— Can occur in the first through the third trimesters. Caused by increased levels of estrogen and progesterone

Relief:

— Wearing a supportive bra.
— Avoiding the use of soap on the nipples and areolar area to prevent drying of skin

➢ **Ankle Edema:**

— Uusually occurs in the second and third trimester. Results from vasodilation, venous stasis, and increased venous pressure below the uterus

Relief:

— Elevating the legs at least twice a day and when resting
— Sleeping in a side-lying position
— Wearing supportive stockings or support hose
— Avoiding sitting or standing in one position for long periods.
— Drinking ample amounts of fluid as allowed.

➢ **Varicose Veins:**

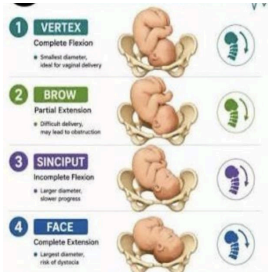
— Usually occur in the second and third trimesters. Result from weakening walls of the veins or valves and venous congestion. Thrombophlebitis is rare, but it may occur.

Relief:

— Wearing supportive stockings or support hose
— Elevating the feet when sitting
— Lying with the feet and hips elevated
— Avoiding long periods of standing or sitting
— Moving about while standing to improve circulation
— Avoiding leg crossing
— Avoiding constricting articles of clothing such as knee-high stockings
— Teaching leg exercises
— Avoiding lying because of sitting position

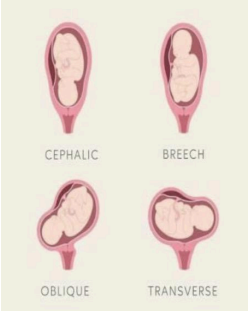
LABOR AND BIRTH

➢ **ATTITUDE**



— Attitude is the relationship of the fetal body parts to one another.
— Normal intrauterine attitude is flexion, in which the fetal back is rounded, the head is forward on the chest, and the arms and legs are folded in against the body.
— The other attitude, extension, tends to present larger fetal diameters.

➢ **FETAL LIE**



— Lie is the relationship between the long (cephalocaudal) axis of the fetal body and the long (cephalocaudal) axis of a woman's body—in other words, whether the fetus is lying in a horizontal (transverse) or a vertical (longitudinal) position.
— Approximately 96% of fetuses assume a longitudinal
— Lie (with their long axis parallel to the long axis of the woman)

➢ **FETAL PRESENTATION**



— Denotes the body part that will first contact the cervix or be born first.
— Or be born first and are determined by the combination of fetal lie and the degree of fetal flexion (attitude).

FALSE CONTRACTIONS VS. TRUE CONTRACTIONS

FALSE	TRUE
Begin and remain irregular.	Begin regularly but become regular and predictable.
Felt first abdominally and remain confined to the abdomen and groin.	Felt first in lower back and sweep around to the abdomen in a wave.

Often disappear with ambulation or sleep.	Continue no matter what the woman's level of activity.
Do not increase in duration, frequency, or intensity.	Increase in duration, frequency, and intensity.
Do not achieve cervical dilatation.	Achieve cervical dilatation

CARDINAL MOVEMENTS (EDFIEREE)

1. ENGAGEMENT

— Engagement is the mechanism whereby the fetus nestles into the pelvis.
— Engagement occurs when the presenting part reaches the level of the ischial spines.

2. DESCENT

— Descent is the process that the fetal head undergoes as it begins its journey through the pelvis.
— Descent is a continuous process, from before engagement until birth, and is assessed by the measurement called station.

3. FLEXION

— Flexion is a process of nodding of the fetal head forward toward the fetal chest.

4. INTERNAL ROTATION

— Internal rotation of the fetus occurs most commonly from the occipitotransverse position, assumed at engagement into the pelvis, to the occipitoanterior position while continuously descending.

5. EXTENSION

— Extension enables the head to emerge when the fetus is in a cephalic position.
— Extension begins after the head crowns.
— Extension is complete when the head passes under the symphysis pubis and occiput, and the anterior fontanel, brow, face, and chin pass over the sacrum and coccyx and are over the perineum.

6. RESTITUTION

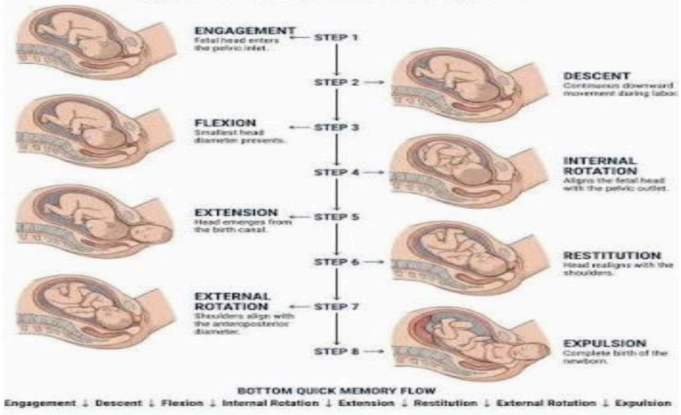
— Restitution is realignment of the fetal head with the body after the head emerges.

7. EXTERNAL ROTATION

— The shoulders externally rotate after the head emerges and restitution occurs so that the shoulders are in the antero-posterior diameter of the pelvis.

8. EXPULSION

— Expulsion is the birth of the entire body.

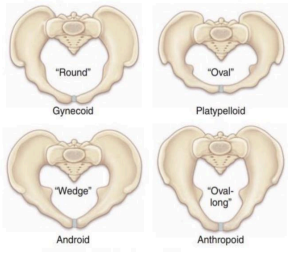


COMPONENTS OF LABOR (5Ps of Labor)

1. PASSAGE

— (Passageway) the pelvis and birth canal
— Refers to the route a fetus must travel from the uterus through the cervix and vagina to the external perineum.
— It is composed of the bony pelvis and soft tissues (cervix, pelvic floor muscles, and vagina)

Types of Pelvis:



— **GYNECOID** – typical female pelvis
— **ANDROID** – is the male or funnel shaped pelvis
— **ANTHROPOID** – long, narrow, and oval shaped
— **PLATYPELLOID** – wide pelvis flattened at brim

Giving Birth Based on Type of Passage

— **Gynecoid** - most favorable type for a vaginal birth
— **Android** - can make labor difficult; baby might move more slowly through the birth canal; may require a C-Section
— **Anthropoid** - may be able to have a vaginal birth, but their labor might last longer.
— **Platypelloid** - baby may have trouble passing; need to have a C-section.

EFFACEMENT AND DILATATION

— **EFFACEMENT** - Thinning and shortening of the cervix
— **DILATATION** –opens completely to 10 centimeters to allow the passage of the baby's head.

2. PASSENGER

— (The fetus) size, position, and presentation.
passenger is the fetus.
— The passenger refers to the baby traveling through the birth canal.
— Its movement depends on several critical fetal variables:

- **Fetal Head Size:** The cranial bones are not fused, allowing them to overlap (molding) to fit through the pelvis.
- **Fetal Lie:** The relationship of the baby's spine to the mother's spine—either longitudinal (parallel, which is ideal) or transverse (horizontal)

- **Fetal Presentation:** The part of the baby entering the pelvis first. Cephalic (head-first) is optimal; breech (buttocks/feet first) or shoulder presentations complicate delivery.
- **Fetal Attitude:** The degree of flexion. A tucked chin (optimal flexion) presents the smallest head diameter, whereas an extended neck makes passage difficult.
- **Fetal Position:** The relationship of the presenting part to the maternal pelvis (e.g., Occiput Anterior, where the baby faces the mother's spine)

3. POWERS

— Power of labor (uterine factors) adequate.
— Powers represent the physiological forces driving the labor process to expel the fetus and placenta.

2 Phases:

1. **Primary Powers (Involuntary):** Uterine contractions.
2. **Secondary Powers (Voluntary):** Maternal pushing efforts.

UTERINE CONTRACTIONS

— Mark of effective uterine contractions is rhythmicity and progressive lengthening and intensity.
— ***Phases of Intensity:***

1. **INCREMENT** - when the intensity of the contraction increases
2. **ACME** - when the contraction is at its strongest
3. **DECREMENT** - when the intensity decreases

— **FREQUENCY** - is the time from the beginning of one contraction to the beginning of the next contraction.
— **DURATION** - It is the time from the beginning of one contraction to the end of the same contraction.

- Between contractions, the uterus relaxes.
- As labor progresses, the relaxation intervals decrease from 10 minutes early in labor to only 2 to 3 minutes.
- Duration of contractions also changes, increasing from 20 to 30 seconds to a range of 60 to 90 seconds.

4. POSITION

— Woman's position during labor. (postures)
— Position refers to the physical posture of the laboring person.
— Frequent position changes and upright postures (walking, sitting, kneeling, squatting) utilize gravity to encourage fetal descent and optimal rotation.
— Changing positions changes the pelvic diameters slightly, allowing the baby more room.

5. PSYCHE

— Woman's psychological outlook
— This is the woman's mental readiness, the emotional and psychological state of the person giving birth.
— The laboring person's mental preparedness, coping mechanisms, and anxiety levels
— **Support system:** Reassurance and a positive environment from nurses, midwives, and partners help reduce stress hormones.

FETAL MONITORING

BIOPHYSICAL PROFILE SCORING

1. **Fetal breathing** - at least one episode of 30secs. of sustained breathing movement w/in 30mins
2. **Fetal movement** - at least 3 episodes of fetal limb or trunk movement w/in 30mins.
3. **Fetal tone** - Observation must extend and then flex extremities or spine at least once in 30 mins.
4. **Fetal heart reactivity** - 2 or more heart accelerations at least 15 beats/min
5. **Amniotic fluid volume** - A range of amniotic fluid between 5 and 25 cm must be present

DAILY FETAL MOVEMENT COUNT (KICKS COUNT)

— 18 – 20 wks. – quickening felt by the mother
— 28 – 38 wks. – 10 x / hr. peaks in intensity

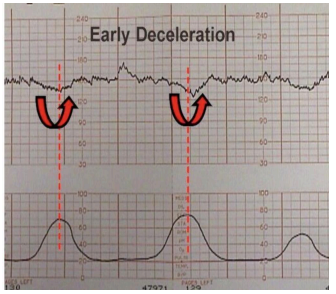
RHYTHM STRIP TESTING

— Use for assessment of the fetal heart rate
— Average FHR – 130 beats/ min.
— Average fetal moves – twice every 10 mins. - causes heart rate to increase

a. **Accelerations**

— Temporary normal increases in FHR caused by fetal movement or compression of the umbilical vein during contraction

b. **Early Decelerations**



— Periodic decreases in FHR resulting from pressure of the fetal head during contractions.
— Beginning when the contractions begins and ends when the contractions end (mirror image)
— Normal – late in labor

c. **Late Decelerations**



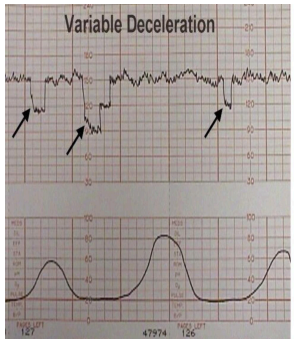
— are a type of fetal heart rate deceleration that can indicate reduced blood flow to the placenta and potential fetal acidemia.

Interventions:

— Stop or slow the administration of oxytocin
— Change the position from supine to lateral to relieve pressure from the Vena Cava
— Administer IVF or O2 as prescribed

— If late decelerations persist – prepare for possible prompt birth of the infant

d. Variable Decelerations



— Decelerations that occur at unpredictable times in relation to contractions.
— Indicate compression of cord - Cord prolapsed
— Fetus is lying on the cord
— Occurs more frequently: - after rupture of membranes
— Oligohydramnios
— U, V or W– shaped waves

Interventions:
— Position: lateral or T-position
— Administer fluids and O2 as prescribed
— If not relieved, amnioinfusion may be prescribed

VEAL CHOP MINE NURSING MNEMONICS

FHR Pattern (VEAL)		Cause (CHOP)		Management (MINE)	
V	Variable deceleration	C	Cord compression	M	Maternal repositioning
E	Early deceleration	H	Head compression	I	Identify labor progress
A	Acceleration	O	Okay!	N	No interventions
L	Late deceleration	P	Placental insufficiency	E	Excute interventions

STAGES OF LABOR AND DELIVERY

STAGES OF LABOR

Labor is traditionally divided into four stages:
1. **First stage of dilatation**, which begins with the initiation of true labor contractions and ends when the cervix is fully dilated;
2. **Second stage**, extending from the time of full dilatation until the infant is born; and
3. **Third or placental stage**, lasting from the time the infant is born until after the delivery of the placenta.
4. **Fourth stage** - first 1 to 4 hours after birth of the placenta, emphasize the importance of the close maternal observation needed at this time.

A. 1ST STAGE – STAGE OF DILATATION

➢ **LATENT PHASE**

— Begins at the onset of regularly perceived uterine contractions and ends when rapid cervical dilatation begins.
— Duration: 6 hours (nullipara); 4.5 hours (multipara)
— Duration of contractions: 20-40 sec
— Interval of contractions: 5-10 minutes
— Cervical dilatation: 0-3cm
— Psyche: excited, still can communicate

➢ **ACTIVE PHASE**

— Cervical dilatation occurs more rapidly
— Duration: 3 hours (nullipara); 2 hours (multipara)
— Duration of contractions: 40 - 60 sec
— Interval of contractions: 3 - 5 minutes
— Cervical dilatation: 4 – 6 cm
— Psyche: frightened, anxious, irritable, but still can comprehend

➢ **TRANSITION PHASE**

— Contractions reach their peak of intensity
— A woman reaches the end of this stage at 10 cm of dilatation.
— Duration of contractions: 60-90 sec
— Interval of contractions: 2-3 minutes
— Cervical dilatation: 8 - 10cm
— Psyche: loss of control

Nursing Care During 1st Stage: DILATATION STAGE

➢ **LATENT AND ACTIVE PHASE**

— Assessments and monitoring
- Physical exam
- Vital signs
- Internal exam
- FHR
- Uterine contractions

Health Teaching

— Bath
— Ambulation
— NPO
— Breathing
— Emptying the bladder
— Sim's Position
— Discourage pushing

Preparation for Birth:

— Perineal Prep
— Perineal Shave
— Administer analgesics as ordered
— Assist in the administration of anesthesia
— Assist in the transport to Delivery Room

➢ **TRANSITION PHASE**

— Comfort Measures
— Proper bearing down techniques

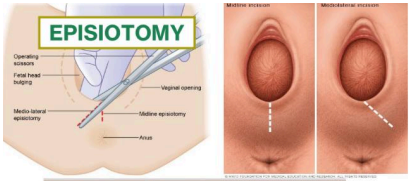
B. 2ND STAGE - STAGE OF EXPULSION

— Period from full dilatation and cervical effacement to birth of the infant; with uncomplicated birth, this stage takes about 1 hour (Archie, 2007).
— A woman feels contractions change from the characteristic crescendo–decrescendo pattern to an overwhelming, uncontrollable urge to push or bear down with each contraction as if to move her bowels.
— Anus may become everted, and stool may be expelled
— As she pushes, using her abdominal muscles to aid the involuntary uterine contractions, the fetus is pushed out of the birth canal.

Nursing Care During 2nd Stage: EXPULSION STAGE

— Put legs at the same time when positioning in lithotomy
— As soon as the head crowns, instruct not to push but to pant

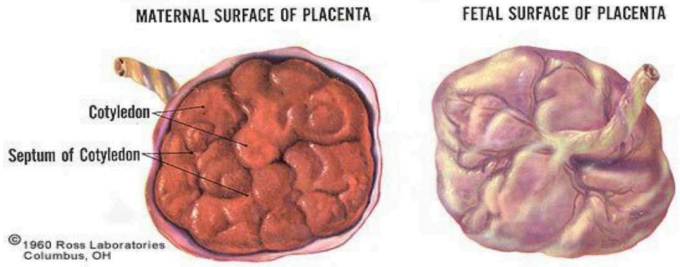
— Assist in episiotomy - a cut (incision) through the area between your vaginal opening and your anus.
— Apply the **Modified Ritgen's Maneuver** (performed to deliver the head)





C. 3RD STAGE - PLACENTA, STAGE

— The placental stage begins with the birth of the infant and ends with the delivery of the placenta.
— Two separate phases are involved: **placental separation and placental expulsion.**
— After the birth of an infant, a uterus can be palpated as a firm, round mass just inferior to the level of the umbilicus.
— After a few minutes of rest, uterine contractions begin again, and the organ assumes a discoid shape.
— It retains this new shape until the placenta has separated, approximately 5 minutes after the birth of the infant.
— Do not hurry the expulsion of the placenta
— Note for the signs of placental separation
— Follow the **Brandt-Andrews Maneuver**
— Take note of the time of placental delivery
— Inspect for the completeness of cotyledons
— Palpate the uterus to determine the degree of contraction
— Administer oxytocin / Methergine after placental delivery
— Inspect for the completeness of cotyledons



Nursing Care During 3rd Stage: PLACENTAL STAGE

— Inspect perineum for lacerations
— Perineal care
— Provide additional blankets
— Allow to sleep to regain lost energy

D. 4TH STAGE - RECOVERY STAGE

— Covers the first 1 to 2 hours after delivery.
— Critical period for monitoring bleeding, uterine contraction, mother's vital signs.
— Medical staff monitor vital signs, bleeding, and uterine cramping while bonding and breastfeeding begin

Nursing Care During 4th Stage: RECOVERY STAGE



Assessments:
— Fundus
— Vital Signs
— Lochia
— Perineum

Health Interventions and Teachings

— Rooming-in concept
— Early ambulation
— Dangling of legs

Post-partum Assessment:

"BUBBLE HE"
— **B** - reast
— **U** - terus
— **B** - owel
— **B** - ladder
— **L** - ochia
— **E** - pisiotomy
— **H** - oman's Sign
— **E** - motional Status

MIDTERMS: CMPA411 (COMPETENCY APPRAISAL 1) - MODULE #7: CARE OF MOTHER WITH COMPLICATIONS OF CHILDBEARING

INCOMPETENT CERVIX (CERVICAL INSUFFICIENCY)

— Incompetent cervix (cervical insufficiency) is a condition characterized by **premature, painless cervical dilation** during the second trimester (commonly 14–20 weeks), resulting from structural or functional weakness of the cervix.
— Often occurs in the **4th or 5th month of pregnancy**
— Leads to recurrent second-trimester pregnancy loss or preterm birth
— *Primary treatment:* **surgical cervical cerclage**

Etiology:
— Structural weakness of cervical tissue
— Congenital cervical abnormalities
— Trauma to the cervix

Risk Factors:
— History of second-trimester miscarriage
— Previous preterm birth without contractions
— Repeated cervical dilation (e.g., D&C procedures)
— Cervical lacerations during childbirth
— Congenital uterine or cervical anomalies
— Exposure to DES (diethylstilbestrol) in utero

Pathophysiology:
— Structural weakness of the cervix
— Inability of cervix to withstand increasing intrauterine pressure
— Progressive cervical shortening and dilation
— Bulging of fetal membranes through the internal os
— Premature rupture of membranes (PROM)
— Pregnancy loss or preterm birth
— *Key feature:* **Dilation occurs without uterine contractions or pain**

Medical Management:
1. Early Identification
— Transvaginal ultrasound to measure cervical length
— Cervical length <25 mm before 24 weeks is concerning

2. Cervical Cerclage
— Performed at **10–14 weeks gestation** (prophylactic)
— A nonabsorbable suture or band is placed around the cervix beneath the mucosa to reinforce the internal os
— *Types:*
- **McDonald procedure** (most common)
- **Shirodkar procedure**

3. Cerclage Removal
— At **37 weeks gestation**
— Or left in place if cesarean birth is planned
— Repeated in future pregnancies if needed.

Pharmacologic Management:
— **Tocolytics** – To suppress uterine contractions if present
— **Progesterone therapy** – May reduce risk of preterm birth
— **Antibiotics** – If infection suspected
— **Analgesics** post-procedure

Nursing Management:
Assessment:
— Vaginal bleeding
— Pelvic pressure
— Visible fetal membranes through cervix
— Absence of pain or contractions
— *Monitor for:*
- Uterine contractions
- Rupture of membranes
- Signs of infection (fever, discharge)

Nursing Interventions:
1. Pre-Cerclage Care
— Provide emotional support (anxiety due to previous losses)
— Encourage bed rest as prescribed
— Maintain hydration
— Administer prescribed tocolytics

2. Post-Cerclage Care
— Monitor fetal heart rate
— Assess for contractions
— Observe for vaginal bleeding
— Monitor temperature for infection
— Assess for leaking of fluid (PROM)
- Fever

3. Client Education
— Avoid:
- Intercourse
- Heavy lifting
- Prolonged standing
— Maintain scheduled prenatal visits
— Report immediately:
- Vaginal bleeding
- Increased uterine contractions
- Rupture of membranes

Expected Outcomes:
— Pregnancy maintained to term
— No infection or rupture of membranes
— Prevention of recurrent second-trimester loss

HYDATIDIFORM MOLE (MOLAR PREGNANCY)

— Hydatidiform mole is a type of gestational trophoblastic disease (GTD) characterized by abnormal proliferation of trophoblastic cells (cells that normally form the placenta).
— The trophoblasts develop abnormally after fertilization.

— The pregnancy results in edematous, grapelike vesicles instead of a viable fetus.
— It may be benign (complete or partial mole) or progress to malignant forms such as choriocarcinoma.

Types:
— **Complete mole** – No fetus present; diffuse trophoblastic proliferation.
— **Partial mole** – Some fetal tissue present but not viable

Etiology:
Abnormal fertilization process:
— **Complete mole:** Fertilization of an empty ovum by one sperm (duplicates) or two sperm.
— **Partial mole:** Fertilization of a normal ovum by two sperm (triploid karyotype).

Risk Factors:
— Maternal age <20 or >35–40 years
— Previous molar pregnancy
— History of miscarriage
— Nutritional deficiencies (low carotene or vitamin A)
— Asian ethnicity (higher incidence reported

Pathophysiology:
— Abnormal fertilization occurs.
— Trophoblastic cells proliferate excessively.
— Chorionic villi become swollen and fluid-filled (hydropic degeneration).
— Formation of grapelike vesicles.
— Markedly elevated **human chorionic gonadotropin (hCG)** levels (>100,000 mIU/mL).
— Excess hCG leads to:
- Hyperemesis gravidarum
- Early-onset preeclampsia (<20 weeks)

— No viable fetus develops (complete mole).
— Risk of persistent GTD or choriocarcinoma if untreated.

Medical Management:
1. Confirm Diagnosis
— Ultrasound: "Snowstorm" or "cluster of grapes" appearance
— Elevated serum hCG levels
— Chest X-ray (to detect metastasis if suspected)

2. Uterine Evacuation
— Vacuum suction curettage (preferred method)
— Oxytocin administered after evacuation to promote uterine contraction

3. Post-Evacuation Monitoring
— Monitor for hemorrhage
— Monitor for infection
— Send tissue for histopathologic evaluation

4. Long-Term Follow-Up
— Serum hCG every 1–2 weeks until normal
— Then every 1–2 months for 1 year
— Prevent pregnancy during follow-up period

Pharmacologic Management:
— **Oxytocin** – To promote uterine contraction post-evacuation
— **Methotrexate** (if persistent GTD or rising hCG) – Chemotherapy agent
— **Actinomycin D** (alternative chemotherapy)
— **Antiemetics** – For hyperemesis
— **Antihypertensives** – If preeclampsia presen

Nursing Management
Assessment:
— Absence of fetal heart tones
— Vaginal bleeding (bright red or dark brown)
— Fundal height larger than expected
— Severe nausea/vomiting
— Early signs of preeclampsia (<20 weeks)
— Monitor hCG level

Nursing Interventions:
1. Pre-Procedure Care
— Provide emotional support (grieving loss of pregnancy)
— Prepare for uterine evacuation
— Baseline vital signs and lab monitoring

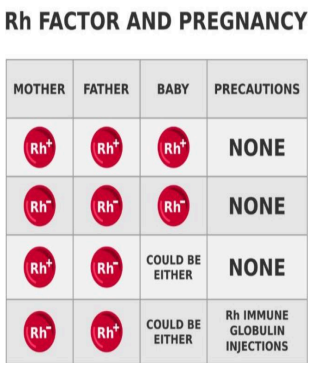
2. Post-Procedure Care
— Monitor for hemorrhage (pad count, vital signs)
— Assess for signs of infection (fever, foul discharge)
— Administer prescribed medications

3. Education
— Importance of regular hCG monitoring
— Use reliable contraception for 1 year
— Avoid pregnancy during follow-up
— Report persistent bleeding or symptoms

4. Psychosocial Support
— Address feelings of loss and anxiety
— Encourage partner involvement
— Provide information about future pregnancy outcome

RH INCOMPATIBILITY

Rh incompatibility occurs when:



— The mother is Rh-negative
— The fetus is Rh-positive
— Fetal blood enters maternal circulation
— The mother forms antibodies against Rh-positive red blood cells.
— This can lead to hemolytic disease of the newborn (HDN) in subsequent pregnancies.

Etiology:
— Maternal exposure to Rh-positive fetal blood
— Formation of anti-D antibodies

Risk Factors:
— Rh-negative mother with Rh-positive partner
— Previous pregnancy with Rh-positive fetus
— Miscarriage or abortion
— Ectopic pregnancy
— Amniocentesis or invasive procedures
— Abdominal trauma during pregnancy

Pathophysiology:
— Rh-negative mother exposed to Rh-positive fetal RBCs
— Maternal immune system produces anti-D antibodies
— In subsequent pregnancy, antibodies cross placenta
— Destruction of fetal RBCs
— Fetal anemia → hyperbilirubinemia
— Severe cases: hydrops fetalis or fetal demise

Medical Management:
— **ABO Typing**
- Determines blood group (A, B, AB, O)

— **Rh Typing**
- Rh-positive → antigen present
- Rh-negative → antigen absent

— **Antibody Screen (Indirect Coombs Test)**
- If Rh-negative & antibody screen negative: Repeat screening and administer Rho(D) immune globulin (RhoGAM) at 28 week

— **Postpartum Management**
- If newborn is Rh-positive → Administer RhoGAM within 72 hours after birth

— **Other Indications for RhoGAM:**
- Miscarriage
- Abortion
- Ectopic pregnancy
- Invasive prenatal procedure

Pharmacologic Management
Rho(D) Immune Globulin (RhoGAM)
— Prevents maternal antibody formation
— Given at:
- 28 weeks gestation
- Within 72 hours postpartum (if infant Rh-positive)
- After pregnancy termination

Nursing Management:
1. Assessment
— Review ABO and Rh typing results
— Monitor antibody screen results
— Assess history of prior sensitization

2. Interventions
— Administer RhoGAM as prescribed
— Verify infant's blood type after delivery
— Educate client on importance of follow-up

3. Client Education
— RhoGAM prevents complications in future pregnancies
— Not harmful to fetus
— Importance of timely administration

PREGNANCY-INDUCED HYPERTENSION (PIH)
— Pregnancy-Induced Hypertension (PIH) refers to hypertension that develops after 20 weeks of gestation in a previously normotensive woman and resolves after delivery.

Major Types:
➢ **Gestational Hypertension**
— BP ≥140/90 mmHg after 20 weeks
— No proteinuria or systemic involvement
➢ **Preeclampsia**
— Hypertension with proteinuria and/or end-organ damage
— *Classified as:*
- Mild preeclampsia
- Severe preeclampsia

➢ **Eclampsia**
— Preeclampsia with seizures
— Life-threatening obstetric emergency
➢ **Chronic Hypertension with Superimposed Preeclampsia**
— Preexisting hypertension worsened by pregnancy

Pathophysiology:
— Abnormal placental implantation leads to **incomplete trophoblastic invasion**
— Placental ischemia causes release of **vasoactive substances**
— *Resulting systemic effects:*
- **Generalized vasospasm**
- ↓ Organ perfusion (kidneys, liver, brain, placenta)
- ↑ Capillary permeability → edema

— *Renal involvement:*
- Glomerular endotheliosis → proteinuria

— *Placental insufficiency:*
- ↓ Oxygen and nutrients → fetal growth restriction

— *In severe cases:*
- Cerebral edema → seizures (eclampsia)
- Hemolysis, elevated liver enzymes, low platelets (HELLP syndrome)

Medical Management
— Early identification through prenatal screening
— Frequent monitoring of:
- Blood pressure
- Urine protein
- Laboratory values (platelets, liver enzymes)

— Fetal surveillance:
- Non-stress test (NST)
- Biophysical profile (BPP)

— Timing of delivery:
- Only **definitive cure** for PIH
- Balances maternal risk vs fetal maturity

Pharmacological Management:
— **Antihypertensives**:

- Methyldopa
- Labetalol
- Hydralazine (IV for severe hypertension)

— **Anticonvulsants**
- **Magnesium sulfate** – drug of choice for seizure prevention

— **Monitor for toxicity:**
- Loss of deep tendon reflexes
- Respiratory depression
- Decreased urine output

— **Corticosteroids**
- Promote fetal lung maturity if preterm delivery anticipate

Surgical Management:
— Cesarean delivery if:
- Severe preeclampsia or eclampsia
- Non-reassuring fetal status
- Failed induction

— Emergency delivery indicated if maternal or fetal condition deteriorates.

Nursing Management:
1. Antenatal (During Pregnancy)
— Monitor BP accurately and consistently
— Assess for warning signs:
- Severe headache
- Visual disturbances
- Epigastric or RUQ pain
- Sudden weight gain or edema

— Monitor urine output and protein
— Educate on:
- Medication compliance
- When to seek emergency care

— Promote rest (left lateral position)
— Continuous fetal monitoring in severe case

2. Intrapartum (During Labor)
— Continuous maternal and fetal monitoring
— Administer magnesium sulfate as ordered
— Maintain seizure precautions:
- Padded side rails
- Suction and oxygen available

— Strict intake and output monitoring
— Limit environmental stimuli
— Prepare for possible emergency deliver

3. Postpartum
— Continue BP monitoring (risk persists postpartum)
— Continue magnesium sulfate for 24 hours post-delivery
— Assess for postpartum hemorrhage
— Educate on follow-up care and future pregnancy risk

GESTATIONAL DIABETES MELLITUS (GDM)

— Gestational Diabetes Mellitus (GDM) is a condition of glucose intolerance with onset or first recognition during pregnancy, usually identified in the 24-28th week of pregnancy.

Categories:
➢ Diet-Controlled GDM (A1)
— Blood glucose controlled through nutrition therapy and exercise
➢ Medication-Controlled GDM (A2)
— Requires insulin or oral hypoglycemic agents to maintain euglycemia

Note: GDM usually resolves after delivery but increases the risk of type 2 diabetes later in life for the mother.

Pathophysiology:
— Pregnancy causes a **diabetogenic state**
— Placenta produces hormones that oppose insulin action:
- Human placental lactogen (hPL)
- Estrogen
- Progesterone
- Cortisol

— These hormones cause **insulin resistance**
— Normal pregnancy: pancreas compensates by ↑ insulin production
— In GDM:
- Inadequate insulin response → **maternal hyperglycemia**

Maternal Effects:
Increased risk of:
— Preeclampsia
— Polyhydramnios
— Birth trauma
— Cesarean delivery

Fetal Effects:
— Excess maternal glucose crosses placenta
— Fetal pancreas responds with **hyperinsulinemia**
— Results in:
- Macrosomia
- Shoulder dystocia
- Neonatal hypoglycemia
- Respiratory distress syndrom

Medical Management
— Universal or risk-based screening (24–28 weeks gestation)
— Diagnostic testing:
- Oral glucose tolerance test (OGTT) (75g)

— Regular monitoring of:
- Fasting blood glucose (<95mg/dL or 5.3 mmol/L)
- Postprandial blood glucose (1 hr pp = <180 mg/dL or 7.8 mmol/L; 2 hrs pp = <153 mg/dL or 6.7 mmol/L)
- Maternal weight gain
- Fetal growth (ultrasound)

— Goal: maintain **euglycemia** to prevent complication

Pharmacological Management:

— **Insulin therapy**
- First-line treatment if diet and exercise are insufficient
- Does not cross the placenta

— **Oral hypoglycemic agents** (used cautiously):
- Metformin
- Glyburide

— Insulin requirements increase as pregnancy progresses

Surgical Management:

— No direct surgical treatment for GDM
— Cesarean delivery may be indicated if:
- Fetal macrosomia
- Failure of labor progression
- Maternal or fetal compromis

Nursing Management:

1. Antenatal (During Pregnancy)

— Monitor blood glucose levels as prescribed
— Teach self-monitoring of blood glucose
— Provide nutrition education:
- Small, frequent meals
- Balanced carbohydrate intake

— Encourage moderate exercise (e.g., walking)
— Assess for signs of hyperglycemia and hypoglycemia
— Monitor fetal well-being (NST, ultrasound)

2. Intrapartum (During Labor)

— Monitor maternal blood glucose levels
— Maintain glucose within target range
— IV insulin and glucose infusion as ordered
— Continuous fetal heart rate monitoring
— Prepare for potential shoulder dystocia or operative delivery.

3. Postpartum

— Monitor maternal blood glucose (usually normalizes)
— Assess neonate for:
- Hypoglycemia
- Respiratory distress

— Encourage breastfeeding (improves glucose metabolism)
— Educate mother on:
- Risk of future type 2 diabetes
- Lifestyle modification
- Need for postpartum glucose testing (6–12 weeks)

HYPEREMESIS GRAVIDARUM

— Hyperemesis Gravidarum (HG) is a severe, persistent form of nausea and vomiting during pregnancy that leads to dehydration, electrolyte imbalance, ketosis, and weight loss, usually occurring in the first trimester but may persist longer.

Categories / Spectrum:

➢ **Mild Nausea and Vomiting of Pregnancy (NVP)**

— Common "morning sickness"
— No dehydration or weight loss

➢ **Moderate Hyperemesis**

— Persistent vomiting
— Mild dehydration and electrolyte imbalance

➢ **Severe Hyperemesis Gravidarum**

— Continuous vomiting
— Weight loss >5% of pre-pregnancy weight
— Ketonuria, dehydration, metabolic imbalance
— Requires hospitalization

Pathophysiology

— Exact cause is unknown; **multifactorial etiology**
— Contributing factors include:
- Elevated **human chorionic gonadotropin** (hCG)
- Increased estrogen levels
- Altered gastric motility
- Psychological stress

— Excessive vomiting leads to:
- **Fluid volume deficit**
- Electrolyte imbalances (↓ sodium, ↓ potassium, ↓ chloride)
- Metabolic alkalosis initially; may progress to acidosis

— Starvation state results in:
- Breakdown of fat → **ketone production**
- Risk for maternal malnutrition

— Severe, prolonged cases may lead to:
- Vitamin B_1 (thiamine) deficiency
- Neurologic complications (e.g., Wernicke encephalopathy)

Medical Management

— Assessment of severity:
- Weight loss
- Urine ketones
- Serum electrolytes

— Outpatient management for mild cases
— **Hospitalization** for moderate to severe HG
— Goals of care:
- Restore hydration
- Correct electrolyte imbalance
- Control vomiting
- Prevent complication

Pharmacological Management

— **Antiemetics**
- Ondansetron
- Metoclopramide
- Promethazine

— **Vitamin therapy**
- Vitamin B_6 (pyridoxine)
- Thiamine supplementation (especially before IV glucose)

— **IV fluids**
- Normal saline or lactated Ringer's
- Add dextrose after thiamine administration

— **Acid-reducing agents (if needed)**
- H_2 blockers or proton pump inhibitor

Surgical Management
— **No definitive surgical treatment**
— Rare and extreme cases:
- Enteral feeding via nasogastric or jejunal tube
- Parenteral nutrition (TPN) if oral intake is impossible

— Pregnancy termination is **not standard management** and considered only as a last resort in life-threatening case.

Nursing Management:
1. Antenatal (During Pregnancy)
— Assess severity of nausea and vomiting
— Monitor:
- Daily weight
- Intake and output
- Urine ketones

— Maintain hydration:
- IV fluids as prescribed

— Administer antiemetics and vitamins
— Provide dietary guidance:
- Small, frequent meals
- Dry, bland foods
- Avoid fatty and spicy foods

— Encourage rest and stress reduction
— Provide emotional support and reassurance

2. Inpatient Nursing Care (Severe Cases)
— Strict intake and output monitoring
— Monitor electrolytes and laboratory values
— Observe for signs of dehydration:
- Dry mucous membranes
- Tachycardia
- Hypotension

— Administer IV medications safely
— Monitor fetal well-being once viable gestation is reached.

3. Discharge Teaching
— Medication adherence
— Gradual reintroduction of oral intake
— When to seek medical help:
- Persistent vomiting
- Inability to tolerate fluids
- Decreased urine output

— Reassure that condition usually improves by mid-pregnancy

PLACENTA PREVIA

— Placenta Previa is a pregnancy complication in which the placenta implants in the lower uterine segment, partially or completely covering the cervical os, leading to painless vaginal bleeding, usually in the second or third trimester.

Types of Placenta Previa:
➢ **Complete (Total) Placenta Previa**
— Placenta completely covers the cervical os
— Vaginal delivery is contraindicated
➢ **Partial Placenta Previa**
— Placenta partially covers the cervical os
➢ **Marginal Placenta Previa**
— Placenta reaches the edge of the cervical os
➢ **Low-Lying Placenta**
— Placenta implanted in lower uterus but does not reach the os
— May resolve as uterus enlarges

Pathophysiology
— Placenta implants abnormally in the **lower uterine segment**
— As pregnancy progresses:
- Lower uterine segment thins and stretches
- Placental attachment site becomes disrupted

— This leads to **separation of placental villi from uterine wall**
— Results in **sudden, painless, bright-red vaginal bleeding**
— Bleeding is maternal in origin
— Risk factors include:
- Previous cesarean birth
- Multiple pregnancies
- Advanced maternal age
- Prior placenta previa
- Smoking

Medical Management:
— Diagnosis primarily via **ultrasound**
— Avoid vaginal examinations
— Management depends on:
- Gestational age
- Severity of bleeding
- Maternal and fetal stability

— Expectant management if:
- Bleeding is minimal
- Pregnancy is preterm

— Planned delivery at **36–37 weeks** if placenta previa persist.

Pharmacological Management
— **Corticosteroids** - Promote fetal lung maturity if preterm delivery anticipated
— **Tocolytics** - May be used short-term to delay labor (if no active bleeding)
— **IV fluids and blood products** - To manage hypovolemia and anemi

Surgical Management
— Cesarean delivery
- Definitive treatment for complete or partial placenta previa
- Emergency cesarean if: Severe hemorrhage, maternal or fetal compromise

— Vaginal delivery only considered in selected marginal or low-lying cases with close monitoring.

Nursing Management
1. Antenatal (During Pregnancy)
— Assess vaginal bleeding:
- Amount, color, onset

— Monitor maternal vital signs
— Monitor fetal heart rate
— Enforce **pelvic rest:**
  - No intercourse
  - No vaginal exams
— Educate patient on:
  - Reporting bleeding immediately
  - Activity restriction
— Prepare for possible hospitalization

2. Intrapartum (During Labor)
— Continuous maternal and fetal monitoring
— Maintain IV access with large-bore catheter
— Prepare blood products if needed
— Monitor for signs of shock
— Prepare for emergency cesarean delivery
— Provide emotional support and reassurance

3. Postpartum
— Monitor for postpartum hemorrhage
— Assess uterine tone and lochia
— Monitor hemoglobin and hematocrit
— Provide education on:
  - Recovery after cesarean birth
  - Future pregnancy risk

ABRUPTIO PLACENTAE (PLACENTAL ABRUPTION)

— Abruptio Placentae is the premature separation of a normally implanted placenta from the uterine wall after 20 weeks of gestation and before delivery, resulting in painful vaginal bleeding and uterine tenderness

Types of Abruptio Placentae:
➢ **Concealed Abruption**
— Blood trapped behind the placenta
— No visible vaginal bleeding
— More dangerous due to hidden hemorrhage
➢ **Revealed Abruption**
— Blood escapes through the cervix
— Visible vaginal bleeding
➢ **Marginal Abruption**
—Separation occurs at placental edge
— Usually less severe bleeding

Pathophysiology:
— Initiating event: rupture of maternal blood vessels in the decidua basalis
— Blood accumulates between placenta and uterine wall → placental separation
— *Separation leads to:*
  - ↓ uteroplacental perfusion
  - Fetal hypoxia and distress
—Uterine muscle becomes irritated and hypertonic
— *Bleeding may be:*
  - External (revealed)
  - Internal (concealed)
— Severe cases may progress to:
— Disseminated Intravascular Coagulation (DIC)
  - Hypovolemic shock

Common Risk Factors:
— Hypertension / preeclampsia
— Abdominal trauma
— Cocaine or smoking
— Previous abruption
— Sudden uterine decompression

Medical Management:
— **Obstetric emergency**
  - Rapid maternal and fetal assessment
  - Continuous fetal heart rate monitoring
— **Assessment of**:
  - Vital signs
  - Vaginal bleeding
  - Uterine tone and tenderness
— **Laboratory evaluation:**
  - Hemoglobin/hematocrit
  - Coagulation profile
— Immediate delivery if:
  - Severe bleeding
  - Fetal distress
  - Maternal instabilit

Pharmacological Management:
— IV fluids for volume replacement
— **Blood and blood products**
  - Packed RBCs
  - Fresh frozen plasma if DIC develops
— **Oxytocin**
  - Used cautiously after delivery to control uterine atony
— Pain management as ordered
— No tocolytics if active bleeding is presen

Surgical Management:
— **Emergency cesarean section**
  - Indicated for fetal distress or uncontrolled hemorrhage
— Vaginal delivery; May be possible if:
  - Fetus is deceased
  - Labor is advanced
  - Maternal condition is stable
— Hysterectomy may be required in catastrophic hemorrhage

Nursing Management:
1. Antenatal / Emergency Care
— Recognize classic signs:
  - Painful vaginal bleeding
  - Rigid, board-like uterus
  - Sudden abdominal pain
— Monitor maternal vital signs frequently
— Continuous fetal heart rate monitoring
— Establish large-bore IV access
— Prepare blood products
— Keep patient NPO
— Provide emotional support to patient and family

2.Intrapartum (During Labor)
— Monitor uterine activity and tone
— Assess for worsening pain or bleeding
— Prepare for emergency delivery
— Assist with rapid stabilization
— Document maternal and fetal responses

3.Postpartum
— Monitor for:
- Postpartum hemorrhage
- Signs of DIC

— Assess uterine tone and lochia
— Monitor laboratory results
— Provide grief support if fetal loss occurs
—Educate regarding recurrence risk in future pregnancies

PREMATURE RUPTURE OF THE MEMBRANES (PROM)

— Premature Rupture of the Membranes (PROM) is the spontaneous rupture of the amniotic membranes before the onset of labor, regardless of gestational age

Etiology:
— Infection (chorioamnionitis)
— Cervical insufficiency
— Overdistended uterus (multiple pregnancy, polyhydramnios)
— Trauma or invasive vaginal procedures
— Smoking
— Previous history of PROM

Pathophysiology:
— Weakening of the amniotic membranes
— Rupture leads to **loss of protective barrier**
— Ascending microorganisms increase risk of infection
— Decreased amniotic fluid compromises fetal environment
— Prolonged rupture increases risk of **maternal and fetal infection**

Nursing Assessment:
— Presence of fluid pooling in vaginal vault
— Positive nitrazine test
— Assess amount, color, consistency, and odor of fluid
— Monitor maternal vital signs (fever = infection)
— Monitor fetal heart rate (tachycardia may indicate infection)
— Assess gestational age

Medical Management:
— Depends on **gestational age** and presence of infection
— Avoid unnecessary vaginal examinations
— Close monitoring of maternal and fetal status
— Expectant management if preterm and no infection
— Plan for delivery if term or if infection develop

Pharmacological Management:
— Antibiotics to prevent or treat infection
— Corticosteroids if preterm to promote fetal lung maturity
— Tocolytics may be considered short-term (if no infection)

Nursing Management:
— Monitor for signs of infection
— Maintain strict aseptic technique
— Limit vaginal examinations
— Monitor fetal heart rate regularly
— Educate patient to report fever, foul-smelling discharge, or contractions
— Provide emotional support

PROLAPSED UMBILICAL CORD

— Prolapsed Umbilical Cord occurs when the umbilical cord slips ahead of or alongside the presenting fetal part, becoming compressed and compromising fetal circulation

Etiology:
— Premature rupture of membranes
— Malpresentation (breech, transverse lie)
— Long umbilical cord
— Multiple gestation
— Unengaged presenting part

Pathophysiology:
— Rupture of membranes allows cord to descend
— Cord becomes compressed between fetus and pelvis
— Reduced blood flow and oxygen to fetus
— Leads to fetal hypoxia and distress

Nursing Assessment:
— Client reports **feeling something** coming out of the vagina
— Visible or palpable umbilical cord
— **Fetal heart rate abnormalities**:
- Bradycardia
- Variable decelerations

— Signs of acute fetal distress

Medical Management:
— Considered an **obstetric emergency**
— Immediate relief of cord compression
— Rapid preparation for **emergency cesarean deliver**

Pharmacological Management:
— Oxygen to mother (8–10 L/min via face mask)
— IV fluids to improve uteroplacental perfusion
— Medications as ordered to support maternal stabilization

Nursing Management:
— Call for immediate assistance
— Elevate presenting part manually to relieve cord pressure
— Position mother in **knee-chest or Trendelenburg position**
— Avoid handling the cord unnecessarily

— Cover exposed cord with sterile saline-soaked gauze
— Continuous fetal heart rate monitoring
— Prepare for emergency cesarean delivery
— Provide emotional support and reassurance

SUPINE HYPOTENSION (VENA CAVA SYNDROME

— Supine hypotension, also called vena cava syndrome, occurs when the **gravid uterus compresses the inferior vena cava and aorta** while the pregnant woman is lying supine, resulting in **decreased venous return, reduced cardiac output, and hypotension**

Etiology:
— Compression of inferior vena cava by enlarged uterus
— Supine positioning during late pregnancy

Pathophysiology:
— Gravid uterus compresses vena cava and aorta
— Decreased venous return to the heart
— Reduced cardiac output
— Decreased maternal blood pressure
— Reduced uteroplacental blood flow → fetal distress

Nursing Assessment:
— Pallor
— Dizziness, faintness, breathlessness
— Tachycardia
— Hypotension
— Sweating; cool, clammy skin
— Signs of fetal distress

Medical Management:
— Immediate maternal repositioning
— Continuous maternal and fetal monitoring

Pharmacological Management:
— None routinely indicated

Nursing Management:
— Position client in *side-lying (left lateral) position*
— Avoid supine positioning
— Place pillow or wedge under hip
— Monitor vital signs
— Monitor fetal heart rate

ABORTION (PREGNANCY LOSS BEFORE 24 WEEKS)

— Abortion is the termination of pregnancy **before 24 weeks' gestation,** either:
- **Spontaneous** (miscarriage)
- **Elective** (induced)

Types of Abortion:
➢ **Miscarriage (Spontaneous)**
— Pregnancy ends naturally.
— *Cervix Status:* May be closed or open
— Most occur in 1st trimester
➢ **Induced**
— Pregnancy intentionally terminated
— *Cervix Status:* Depends on method
— May be therapeutic or elective
➢ **Threatened**
— Spotting and cramping but pregnancy may continue
— *Cervix Status:* Closed
— Monitor and encourage rest
➢ **Inevitable**
— Bleeding and cramping; pregnancy cannot continue
— *Cervix Status:* Open (dilating/effacing)
— Prepare for pregnancy loss
➢ **Incomplete**
— Some products expelled, some retained
— *Cervix Status:* Open
— Risk for bleeding and infection
➢ **Complete**
— All products expelled
— *Cervix Status:* Cervix closing
— Bleeding decreases
➢ **Missed**
— Fetal death but tissue retained
— *Cervix Status:* Closed
— Risk for DIC if retained long
➢ **Habitual**
— ≥3 consecutive miscarriages
— *Cervix Status:* Varies
— Needs further evaluation

Risk Factors:
— Advanced maternal age
— Previous miscarriage
— Previous elective abortion
— Uterine abnormalities (adhesions,fibroids)
— Prolonged time to achieve pregnancy
— Low serum progesterone
— Celiac disease
— Polycystic ovarian syndrome (PCOS)
— Thyroid dysfunction or Cushing's syndrome
— Systemic lupus erythematosus (SLE)
— Infection, fever, trauma
— Low BMI (<18.5)
— Smoking, alcohol, cocaine use
— High caffeine intake
— Certain medications

***Assessment*:**
A. Clinical Findings
— Spontaneous vaginal bleeding
— Lower abdominal cramping or contractions
— Passage of blood clots or tissue
— Open or closed cervix (depends on type)

B. Complications
— Excessive bleeding → hemorrhage
— *Signs of shock:*
- Hypotension
- Tachycardia
- Pallor
- Weak pulse

Medical Management:
— Ultrasound to confirm viability
— Serial hCG monitoring
— Expectant management (if stable)
— Dilation and curettage (D&C) for incomplete abortion
— Misoprostol for medical management (as prescribed)

Pharmacologic Management:
— **Misoprostol** – Uterine evacuation
— **Oxytocin** – Control bleeding
— **Rho(D) immune globulin** – For Rh-negative clients
— **Analgesics** – Pain control
— **IV fluids** – If hemorrhage present

Nursing Management:
A. Monitoring
— Maintain bed rest (if prescribed)
— Monitor vital signs
— Assess amount, color, and characteristics of bleeding
— Monitor cramping

B. Evaluation of Blood Loss
— Count and weigh perineal pads
- 1 gram = 1 mL blood loss
— Save expelled tissue and clots for examination

C. Prevention of Complications
— Maintain IV fluids as prescribed
— Monitor for hemorrhage and shock
— Prepare for D&C in incomplete abortion
— Administer Rho(D) immune globulin to Rh-negative client

D. Psychological Support
— Encourage verbalization of feelings
— Acknowledge grief
— Provide privacy
— Include partner/family in support
— Offer referral to counseling if needed

PRETERM LABOR

— Preterm labor is labor that occurs **after 24 weeks but before 37 weeks of gestation,** resulting in cervical changes with or without membrane rupture

Etiology:
— Infection
— History of medical or obstetric complications
— Substance abuse
— Multifetal pregnancy
— Uterine overdistention
— Anemia
— Maternal age <18 or >40
— First pregnancy

Pathophysiology:
— Uterine irritability or overstimulation
— Cervical shortening and dilation
— Premature activation of labor process
— Possible membrane rupture

Nursing Assessment:
— Uterine contractions (painful or painless)
— Abdominal cramping ± diarrhea
— Low back pain
— Pelvic pressure or heaviness
— Change in vaginal discharge (amount, color, odor)
— Ruptured membranes
— Positive fetal fibronectin
— Shortened cervical length

Medical Management:
— Identification and treatment of infection
— Activity restriction
— Hydration
— Bed rest with lateral positioning
— Continuous fetal monitoring

Pharmacological Management:
— **Tocolytics** to suppress labor
— **Corticosteroids** for fetal lung maturity
— **17 alpha-hydroxyprogesterone caproate (17P)** to reduce recurrence risk

Nursing Management:
— Monitor uterine activity
— Monitor fetal heart rate
— Administer IV fluids
— Monitor for medication side effects
— Educate client on preterm labor signs
— Encourage rest and hydration

PRECIPITOUS LABOR AND DELIVERY

— Precipitous labor is labor and delivery that lasts less than 3 hours from onset of contractions to birth

Etiology:
— Strong uterine contractions
— Multiparity
— Low tissue resistance

Pathophysiology:
— Rapid cervical dilation
— Forceful uterine contractions
— Increased risk of maternal and fetal injury

Medical Management:
— Immediate preparation for delivery
— Support safe, controlled birth

Pharmacological Management:
— None routinely indicated
— Nursing Assessment
— Rapid, frequent contractions
— Minimal time between contractions
— Sudden urge to push
— Signs of imminent birth

Nursing Management:
— Prepare precipitous delivery tray
— Stay with the client at all times

— Provide emotional support
— Encourage panting between contractions
— Do NOT attempt to stop delivery
— Perform emergency delivery steps if provider unavailable
— Maintain warmth and safety of newborn

DYSTOCIA

— Dystocia refers to **difficult, prolonged, or painful labor** caused by problems with uterine contractions, the fetus, or the maternal pelvis

Etiology:
— Hypotonic uterine contractions
— Hypertonic uterine contractions
— Fetal macrosomia
— Malposition or malpresentation
— Pelvic abnormalities

Pathophysiology:
— Ineffective or excessive uterine contractions
— Failure of fetal descent
— Prolonged labor
— Increased risk of maternal exhaustion and fetal compromise

Medical Management:
— Pelvic assessment and imaging
— Amniotomy if indicated
— Oxytocin for hypotonic contractions
— Possible operative delivery

Nursing Assessment:
— Abnormal contraction patterns
— Excessive abdominal pain
— Lack of labor progress
— Maternal or fetal tachycardia
— Fetal distress

Pharmacological Management:
— Oxytocin (for hypotonic contractions only)
— Analgesics and sedatives as prescribed
— Antibiotics if infection risk present

Nursing Management:
— Monitor fetal heart rate continuously
— Monitor uterine contractions
— Maintain hydration (IV fluids)
— Change maternal position frequently
— Monitor amniotic fluid color
— Provide comfort measures
— Teach breathing and relaxation techniques
— Assess for cord prolapse after membrane rupture

INTRAUTERINE FETAL DEMISE (IUFD)/FETAL DEATH IN UTERO

— Intrauterine Fetal Demise (IUFD) refers to the **death of a fetus after 24 weeks of gestation and before birth.**
— Also called **fetal death in utero** or stillbirth (if delivered).
— A serious complication if the deceased fetus is retained for **3–4 weeks or longer.**

Etiology:
— Placental insufficiency
— Umbilical cord accidents
— Maternal hypertension or diabetes
— Infection (TORCH infections)
— Trauma
— Rh incompatibility
— Congenital anomalies

Risk Factors:
— Advanced maternal age
— Poorly controlled chronic illness
— Substance abuse
— Previous IUFD
— Multiple gestation

Pathophysiology:
— Fetal death occurs in utero.
— Placental circulation ceases.
— Fetal tissues release thromboplastin into maternal circulation.
— Activation of clotting cascade.
— Excessive clot formation → depletion of clotting factors.
— Development of DIC:
— Widespread microclots
— Severe bleeding due to consumption of clotting factor

Assessment;
A. Maternal Findings
— Loss of fetal movement
— Absence of fetal heart tones
— Decrease in fundal height
— Maternal weight loss
— No fetal cardiac activity on ultrasound

B. Ultrasound Findings
— No fetal heartbeat
— Overlapping fetal skull bones (Spalding sign)
— Other findings suggestive of fetal demise

C.Laboratory Findings (DIC Risk)
— Low hemoglobin and hematocrit
— Low platelet count
— Prolonged PT and aPTT
— Elevated fibrin degradation products
— Bleeding from IV or puncture sites

Medical Management:
— **Confirmation of Diagnosis**
  - Ultrasound confirmation
  - Fetal monitoring
— **Delivery of the Fetus**
  - Induction of labor (usually preferred)
  - Cesarean birth if indicated
— **Monitoring for Complications**

- Assess for DIC
- Monitor coagulation profile

— **Management of DIC (if occurs)**
- IV fluids
- Blood transfusions
- Fresh frozen plasma
- Platelets
- Cryoprecipitate

Pharmacologic Management:
— **Oxytocin** – To induce labor
— **Prostaglandins** – Cervical ripening
— **Broad-spectrum antibiotics** – If infection suspected
— **Blood and blood products** – If DIC develops
— **Analgesics** – Pain management

Nursing Management:
— Education and Follow-Up
— Explain cause if known
— Discuss future pregnancy planning
— Refer to grief counseling or support groups
— Provide written discharge instructions
— Encourage postpartum follow-up

Emotional and Psychosocial Care:
— Encourage expression of grief
— Accept responses such as sadness, anger, guilt, or hostility
— Offer opportunity to see and hold the baby (if desired)
— Provide memory-making options (photos, footprints, keepsakes)
— Respect religious, spiritual, and cultural practices
— Include partner and family in care

Expected Outcomes:
— Safe delivery without maternal complications
— Early identification and treatment of DIC
— Emotional support and healthy grieving process
— Appropriate referral for long-term coping support

POST PARTUM HEMORRHAGE (PPH) AND SHOCK.

— Postpartum Hemorrhage (PPH) is excessive bleeding after childbirth defined as:
- **> 500 mL** after vaginal delivery
- **> 1000 mL** after cesarean delivery

— OR **≥10% drop in hemoglobin/hematocrit** from admission
— With signs and symptoms of hypovolemia

Classification:
— **Early PPH:** Within the first 24 hours postpartum
— **Late PPH:** After 24 hours postpartum
- Greatest risk period:

— First 4 hours after delivery
— Followed by the first 24 hours

Etiology (Causes):
A.Primary Causes
— Uterine atony (most common)
— Cervical or vaginal lacerations
— Hematoma (cervix, perineum, labia)
— Retained placental fragments
— Obesity

Pathophysiology:
— Uterus fails to contract effectively (atony)
— Open blood vessels at placental site remain open
— Excessive blood loss occurs
— Decreased circulating blood volume
— Compensatory tachycardia
— Hypotension and poor tissue perfusion
— If untreated → Hypovolemic shock

Assessment:
A. Bleeding
— Perineal pad soaked within 15 minutes
— Continuous trickling of blood
— Large clots

B. Quantification of Blood Loss
— Weigh pads and linens
— Subtract dry weight
- **1 gram = 1 mL blood loss**

C. Early Signs of Shock
— Restlessness
— Increased pulse rate (tachycardia)
— Rapid, shallow respirations

D. Late Signs
— Decreased blood pressure
— Cool, clammy skin
— Pale/ashen/gray appearance
— Decreased urine output(<30 mL/hr)
— Altered level of consciousness
— Weakness, light-headedness, dyspnea

Medical Management:
— Immediate fundal massage
— Establish large-bore IV access
— Rapid IV fluid replacement
— Oxygen administration
— Blood transfusion if needed
— Surgical intervention if bleeding persists:
- Uterine tamponade
- Uterine artery ligation
- Hysterectomy (last resort)

Pharmacologic Management:
— **Oxytocin (Pitocin)** – First-line uterotonic
— **Methylergonovine** (Methergine) – Avoid in hypertension
— **Carboprost (Hemabate)** – Avoid in asthma
— **Misoprostol (Cytotec)**
— **Tranexamic acid (TXA)**
— **Blood and blood products** (PRBCs, FFP, platelets)

Nursing Management:
A. Immediate Interventions

— Stay with the client
— Call for help
— Massage fundus (if boggy)
— Assess uterine tone and position
— Monitor vital signs every 5–15 minutes

B. Ongoing Monitoring

— Measure and weigh blood loss
— Monitor urinary output (insert Foley catheter if ordered)
— Assess level of consciousness
— Monitor laboratory results (Hgb, Hct, coagulation profile)

C. Prevent Shock

— Maintain IV fluids
— Administer uterotonics as prescribed
— Administer oxygen (8–10 L/min via face mask)
— Prepare for possible transfusion

D. Emotional Support

— Reassure client and family
— Explain interventions
— Provide calm environment

COMMONLY USED MEDICATIONS

Generic Name	Brand Name	Indication (OB Use)	Mechanism of Action	Before Giving	When Giving	After Giving
Nifedipine	Adalat, Procardia	• Preterm labor (tocolytic) • Hypertension in pregnancy	Calcium channel blocker → relaxes uterine smooth muscle & lowers BP	• Check BP (hold if SBP <90 mmHg) • Assess FHR & contractions • Check cardiac history	• Monitor BP q15–30 mins • Monitor pulse (tachycardia) • Watch dizziness/flushing	• Continue BP monitoring • Assess fetal status • Monitor for pulmonary edema
Methyldopa	Aldomet	• Chronic hypertension in pregnancy	Central alpha-2 agonist → decreases sympathetic outflow → lowers BP	• Baseline BP • Check liver function • Assess depression history	• Monitor BP • Observe sedation & orthostatic hypotension	• Monitor liver enzymes • Assess for postpartum depression
Oxytocin	Pitocin	•Induction/ augmentation of labor • Prevention & treatment of postpartum hemorrhage	Stimulates uterine smooth muscle → increases contraction strength & frequency	• Baseline FHR • Assess cervical dilation • Rule out CPD/fetal distress	• Continuous FHR monitoring • Monitor contractions (tachysystole risk) • Use IV pump	• Assess uterine tone • Monitor lochia • Monitor I&O (water intoxication)
Dexamethasone	Decadron	• Promote fetal lung maturity in threatened preterm labor (24–34 weeks)	Corticosteroid → stimulates surfactant production in fetal lungs	• Confirm GA • Check maternal glucose • Assess infection	• Monitor blood glucose • Observe for infection	• Continue glucose monitoring • Monitor fetal status

Generic Name	Brand Name	Indication (OB Use)	Mechanism of Action	Before Giving	When Giving	After Giving
Misoprostol (abortion)	Cytotec	• Medical abortion • Cervical ripening • Management of missed/incomplete abortion	Prostaglandin E1 analog → cervical softening & uterine contractions	• Confirm pregnancy & GA • Baseline VS • Check hemoglobin	• Monitor bleeding • Assess pain • Monitor uterine activity	• Monitor for excessive bleeding • Confirm expulsion • Provide emotional support
Methylergonovine	Methergine	• Prevention & treatment of postpartum hemorrhage	Ergot alkaloid → sustained uterine contraction	• Check BP (contraindicated in HTN) • Assess uterine tone	• Monitor BP closely • Assess uterine firmness	• Monitor lochia • Watch for hypertension, chest pain
Indomethacin	Indocin	• Tocolytic for preterm labor (<32 weeks only)	NSAID → inhibits prostaglandin synthesis → decreases uterine contractions	• Confirm GA (<32 weeks) • Assess renal function • Check amniotic fluid	• Monitor contractions • Monitor FHR	• Monitor for oligohydramnios • Assess for premature ductus arteriosus closure
Magnesium Sulfate	$MgSO_4$	• Prevention & control of seizures in preeclampsia/eclampsia • Neuroprotection in preterm birth • Tocolytic (secondary use)	CNS depressant; blocks neuromuscular transmission & decreases acetylcholine release → prevents seizures	• Baseline BP, RR (>12/min) • Check DTRs (should be present) • Urine output (>30 mL/hr) • Ensure calcium gluconate available (antidote)	• Monitor RR (hold if <12/min) • Monitor DTRs (loss = toxicity) • Monitor urine output • Continuous FHR monitoring	• Continue monitoring for toxicity • Assess for magnesium toxicity (↓RR, absent reflexes, ↓UO) • Monitor newborn for respiratory depression

MIDTERMS: CMPA411 (COMPETENCY APPRAISAL 1) - MODULE #8: CARE OF THE NEWBORN (PEDIA WELL)

INITIAL CARE OF THE NEWBORN

A. ASSESSMENT

1. Observe or assist with initiation of respirations.
2. Assess Apgar score.
3. Note characteristics of cry.
4. Monitor for nasal flaring, grunting, retractions, or abnormal respirations, such as a seesaw respiratory pattern (rise and fall of the chest and abdomen do not occur together).
5. Assess for central cyanosis and acrocyanosis.
6. Obtain vital signs.
7. Observe the newborn for signs of hypothermia or hyperthermia.
8. Assess for gross anomalies.I. INITIAL CARE OF THE NEWBORN

B. INTERVENTIONS

1. Suction the mouth first and then the nares if needed with a bulb syringe.
2. Dry the newborn and stimulate crying by rubbing the back.
3. Maintain temperature stability; wrap the newborn in warm blankets and place a stockinette cap on the newborn's head.
4. Keep the newborn with the birthing parent to facilitate bonding.
5. Place the newborn at the birthing parent's breast/ chest if breast-feeding/chest-feeding is planned, or place the newborn on the birthing parent's chest for skin-to-skin contact.
6. Place the newborn in a radiant warmer.

APGAR SCORING

TABLE 28.1 Five Vital Indicators of Apgar Scoring

Indicator	0 Points	1 Point	2 Points
Heart rate	Absent	<100 beats per minute	≥100 beats per minute
Respiratory rate and effort	Absent	Slow, irregular breathing; weak cry	Good rate and effort; vigorous cry
Muscle tone	Flaccid, limp	Minimal flexion of extremities	Good flexion, active motion
Reflex irritability	No response	Minimal response (grimace) to suction or to gentle slap on soles	Responds promptly with a cry or active movement
Skin color	Pallor or cyanosis	Body skin color normal, extremities blue	Body and extremity skin color normal

TABLE 28.2 Apgar Score Interventions

Score	Intervention
8-10	No intervention required except to support newborn's spontaneous efforts
4-7	Stimulate; rub newborn's back; administer oxygen to newborn; rescore at specific intervals
0-3	Newborn requires full resuscitation; rescore at specific intervals

INITIAL PHYSICAL EXAMINATION

A. GENERAL GUIDELINES

1. Keep the newborn warm during the examination.
2. Begin with general observations; then perform assessments that are least disturbing to the newborn first. Initiate nursing interventions for abnormal findings, and document findings.
3. The Ballard Scale may be used for gestational age assessment; in this scale, scores are assigned to physical and neurological criteria.

B. VITAL SIGNS

➢ **Heart Rate (Resting)**

— 110 to 160 beats per minute (apical), 90 to 110 beats per minute (if sleeping), up to 180 beats per minute (if crying)

— Auscultate at the fourth intercostal space for 1 full minute to detect abnormalities.

➢ **Respirations**

— 30 to 60 breaths per minute.

— Assess for 1 full minute.

➢ **Heart and Respiratory Rate Order**

— Assess heart rate and respiratory rate before other vital signs when the newborn is resting or sleeping.

➢ **Axillary Temperature**

— 97.7° F (36.5° C) to <99.5° F (37.5° C)

— Measure axillary temperature.

➢ **Blood Pressure**

— 80 to 90/40 mmHg (not usually done in term newborn unless cardiac issue suspected)

— Not typically done unless cardiac issues are suspected.

C. BODY MEASUREMENTS (APPROXIMATE)

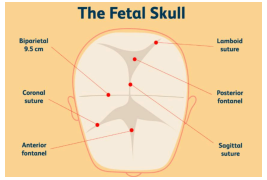
— Length - 18 to 22 inches (45 to 55 cm)

— Weight - 2500 to 4500 g (5.5 to 5.75 lbs)

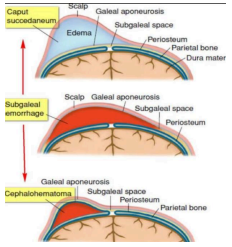
— Head Circumference- 33 to 35 cm (13.2 to 14 inches)

D. HEAD

1. Head should be one-fourth of the body length (cephalocaudal development).
2. Bones of the skull are not fused.
3. Sutures (connective tissue between the skull bones) are palpable and may be overlapping because of head molding but should not be widened.



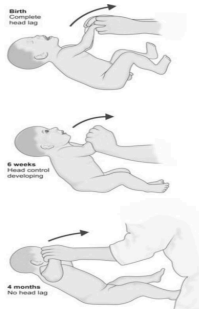
4. Fontanels are unossified membranous tissue at the junction of the sutures. The size of fontanels can vary because of molding and individual differences.
5. Molding is asymmetry of the head resulting from pressure in the birth canal; molding disappears in about 72 hours.
6. ***Masses from birth trauma:***



a. **Caput succedaneum** is edema of the soft tissue over bone (crosses over suture line); it subsides within a few days.

b. **Cephalhematoma** is swelling caused by bleeding into an area between the bone and its periosteum (does not cross over suture line); it usually is absorbed within 6 weeks with no treatment.

7. ***Head lag:***



a. Common when pulling the newborn to a sitting position.

b. When prone, the newborn should be able to lift the head slightly and turn the head from side to side.

c. c. After 6 months old, significant head lag indicates cerebral injury and further evaluation is necessary.

E. EYES

1. Slate-gray (light skin), dark-blue, or brown-gray (dark skin)
2. Symmetrical and clear.
3. Pupils equal, round, react to light and accommodation
4. Blink reflex present
5. Eyes cross because of weak extraocular muscles
6. Ability to track and fixate momentarily
7. Red reflex present
8. Eyelids often edematous as a result of pressure during the birth process and the effects of eye medication

c.

F. EARS

1. Symmetrical
2. Firm cartilage with recoil
3. Top of pinna on or above line drawn from outer canthus of eye.
4. **Low-set ears** associated with Down's syndrome, renal anomalies, or other genetic or chromosomal syndromes

G. NOSE

1. Flat, broad, in center of face
2. Obligatory nose breathing
3. Occasional sneezing to remove obstructions
4. Nares are patent and should not flare (flaring is an indication of respiratory distress)

H. MOUTH

1. Pink, moist gums
2. Soft and hard palates intact
3. Epstein's pearls (small, white cysts) may be present on a hard palate.
4. Uvula in midline
5. Freely moving tongue, symmetrical, has short frenulum
6. Sucking and crying movements symmetrical
7. Able to swallow
8. Root and gag reflexes present

I. NECK

1. Short and thick
2. Head held in midline
3. Trachea midline
4. Good range of motion and ability to flex and extend
5. Assess for torticollis (head inclined to one side as a result of contraction of muscles on that side of the neck)

J. CHEST

1. Circular appearance because anteroposterior and lateral diameters are about equal (approximately **30 to 33** cm [12 to 13.2 inches] at birth)
2. Diaphragmatic respiration—chest and abdomen should rise and fall in synchrony, not in a seesaw pattern
3. Bronchial sounds heard on auscultation
4. Nipples prominent and often edematous; milky secretion (witch's milk) common
5. Breast tissue present
6. Clavicles need to be palpated to assess for fractures.

K. SKIN

1. Pinkish red (light-skinned newborn) to pinkish brown or pinkish yellow (dark-skinned newborn)
2. Nail beds pink, with transient cyanosis immediately after birth
3. **Vernix caseosa**, a cheesy white substance, on entire body in preterm newborns, but is more prominent between folds closer to term; may be absent after 42 weeks of gestation
4. **Lanugo**, fine body hair, might be seen, especially on the back.
5. **Milia**, small white sebaceous glands, appear on the forehead, nose, and chin.
6. Dry, peeling skin, increased in postmature newborns
7. Dark-red **(plethoric)** color is common in premature newborns.
8. **Cyanosis** may be noted with hypothermia, infection, and hypoglycemia and with cardiac, respiratory, or neurological abnormalities.
9. **Acrocyanosis** (peripheral cyanosis of hands and feet) is normal in the first few hours after birth and may be noted intermittently for the next 7 to 10 days
10. Assess for **ecchymosis and petechiae** resulting from trauma of birth.
11. Check for **Mongolian spots.**
12. Check for edema around eyes, face, legs, dorsa of hands, feet, and scrotum or labia.
13. Assess skin turgor over the abdomen to determine hydration status.
14. Observe for forceps marks.
15. **Harlequin sign:**
    a. Deep pink or red color develops over one side of the newborn's body while the other side remains pale or of normal color.
    b. Harlequin sign may indicate shunting of blood that occurs with a cardiac problem or may indicate sepsis.

L. ABDOMEN

1. **Umbilical Cord**

— Normal: 3 blood vessels — 2 arteries and 1 vein.

— **Two-vessel cord:** 1 artery and 1 vein; associated with increased risk of IUGR and genetic/chromosomal abnormalities.
— **Small, thin cord:** May indicate poor fetal growth.
— **Nursing action:** Notify the PHCP if fewer than 3 vessels are identified.
— **Cord integrity:** Check that the cord is intact and the clamp is secure.
— **Clamp removal:** Keep the clamp secured until the cord is dry and occluded, with no bleeding; follow hospital protocol.
— **Bleeding/drainage:** Assess and report any bleeding or abnormal discharge.
— **Cord care:** Follow hospital protocol and PHCP instructions for cleansing.
— **Signs of infection:** Moistness, oozing, discharge, and redness at the base require prompt assessment and provider notification; antibiotics may be prescribed.

2. Gastrointestinal

— Monitor cord for meconium staining.
— Assess for umbilical hernia.
— Assess for abdominal depression associated with diaphragmatic hernia.
— Assess for abdominal distention associated with obstruction, mass, or sepsis.
— Monitor bowel sounds (present within the first hour after birth).

3. **Anus**

— Ensure that the anal opening is present.
— First stool meconium should pass within first 24 hours (may be delayed in premature births)

M. GENITALS

1. Newborn with a Vagina

— Labia may be swollen; clitoris may be enlarged.
— Smegma may be present (thick, white mucous discharge).
— Pseudomenstruation, caused by the withdrawal of the hormone estrogen, is possible (blood-tinged mucus).
— Hymen tag may be visible.
— First voiding should occur within 24 hours.

2. Newborn with a Penis

— Prepuce (foreskin) covers glans penis.
— Scrotum may be edematous, pendulous, and covered with rugae.
— Verify meatus at tip of penis.
— Testes are descended but may retract with cold.
— Assess for hernia or hydrocele; a hernia is easier to detect when the newborn is crying.
— First voiding should occur within 24 hours.

N. SPINES

1. Straight
2. Posture flexed
3. Supportive of head momentarily when prone
4. Chin flexed on upper chest
5. Well-coordinated, sporadic movements
6. A degree of hypotonicity or hypertonicity may indicate central nervous system damage.
7. Assess for hair tufts and dimples along the spinal column (may be indicative of a possible opening).

O. EXTREMITIES

1. Flexed
2. Full range of motion; symmetrical movements and equal muscle tone
3. Fists clenched
4. Ten fingers and 10 toes, all separate
5. Legs bowed
6. Major gluteal folds even
7. Creases on soles of feet
8. Assess for fractures (especially clavicle) or dislocations (hip) from birth trauma.
9. Assist PHCP to assess for developmental dysplasia of the hip; when thighs are rotated outward, no clicks should be heard (**Ortolani's sign and Barlow's sign** are the two assessment tools for developmental dysplasia of the hip).
10. Pulses palpable (radial, brachial, femoral)

BODY SYSTEMS ASSESSMENT AND INTERVENTIONS

A. CARDIOVASCULAR SYSTEM

1. Keep the newborn warm.
2. Measure the apical heart rate for 1 full minute.
3. Listen for murmurs; assess oxygen saturation via pulse oximetry if a murmur is heard.
4. Palpate pulses.
5. Assess for cyanosis; blanch the skin on the trunk and extremities to assess circulation.
6. Observe cardiac distress when the newborn is feeding.

B. RESPIRATORY SYSTEM

1.Suction the airway as necessary: Use a bulb syringe for upper airway suctioning (compress bulb before insertion) and a French catheter for deeper suctioning
2. Observe respiratory distress and hypoxemia.
   a. Abnormal breath sounds
   b. Nasal faring
   c. Increasingly severe retractions
   d. Grunting
   e. Pallor
   f. Cyanosis
   g. Bradycardia and periods of apnea lasting longer than 15 seconds
3. Administer oxygen if necessary and as prescribed.

C. HEPATIC SYSTEM

1. **Normal or physiological jaundice** appears after the first 24 hours in full-term newborns and after the first 48 hours in premature newborns; jaundice occurring before this time (pathological jaundice) may indicate early hemolysis of red blood cells and must be reported to the PHCP.

2. Physiological jaundice peaks on about the fifth day of life (indirect bilirubin levels 6 to 7 mg/dL [90 to 105 mcmol/L]).
3. Feed early to stimulate intestinal activity and to keep the bilirubin level low.
4. Prevent chilling because hypothermia can cause acidosis that interferes with bilirubin conjugation and excretion.
5. The liver stores the iron passed from the birthing parent for 5 to 6 months.
6. Glycogen storage occurs in the liver.
7. The newborn is at risk for hemorrhagic disorders; coagulation factors synthesized in the liver depend on vitamin K, which is not synthesized until intestinal bacteria are present.
8. Handle the newborn carefully, and monitor for any **bruising or bleeding episodes.**
9 Watch for **meconium stools** and subsequent stools.
10. Administer intramuscular dose of **phytonadione** to the newborn as prescribed to prevent hemorrhagic disorders (usually 0.5 to 1 mg is prescribed); administer in lateral aspect of the middle third of the vastus lateralis muscle on the left leg
11. Assess the newborn's **hemoglobin and blood glucose levels.**

D. RENAL SYSTEM

1. The immature kidneys are unable to concentrate urine.
2. A **weight loss of 5% to 10%** during the first week of life occurs as a result of water loss and limited intake; birth weight should be regained by 10 to 14 days after birth.
3. Weigh the newborn daily.
4. Monitor intake and output; weigh diapers if necessary **(1 g of diaper weight = 1 mL of urine).** By the fourth day of life, six wet diapers can be expected; parents should be informed that six wet diapers indicate the infant is taking in adequate fluid.
5. If the diaper requires weighing, record the weight before putting it on the newborn; after the newborn voids, reweigh the diaper and subtract the prevoided weight.
6. Assess for signs of dehydration (dry mucous membranes, sunken eyeballs, poor skin turgor, sunken fontanels).

E. IMMUNE SYSTEM

1. Newborn receives passive immunity via the placenta (immunoglobulin G).
2. Newborn receives passive immunity from colostrum (immunoglobulin A).
3. Elevations in immunoglobulin M indicate infection in utero.
4. Use aseptic technique and standard precautions when caring for the newborn.
5. Ensure meticulous handwashing.
6. Ensure that an infection-free staff cares for the newborn.
7. Monitor the newborn's temperature.
8. Observe any cracks or openings in the skin.
9. Administer eye medication within 1 hour after birth to prevent ophthalmia neonatorum. A chemical conjunctivitis can occur within 24 hours of instillation of ophthalmic prophylaxis; symptoms include mild lid edema and a sterile, nonpurulent eye discharge, which need to be reported.
10. *Provide cord care:*
    - Teach the parent how to perform cord care.
    - Keep the cord clean and dry; alcohol wipes may be prescribed for cleaning the cord only if it becomes soiled.
    - Keep the diaper from covering the cord; fold the diaper below the cord.
    - Assess the cord for odor, edema, or discharge.
    - The newborn is typically washed via a sponge bath until the cord falls off, follow alternate instructions if provided by PHCP.
11. *Circumcision Care:*
    - Healthy newborn only; delay if preterm/sick.
    - Petroleum jelly gauze → avoid with PlastiBell.
    - Remove gauze after first voiding.
    - Keep diaper loose.
    - Observe for edema, infection, bleeding.
    - Teach parents proper site care.
    - Clean with warm water after each voiding.
    - Milky covering = normal; do not remove.
    - Monitor for urinary retention.
    - Newborn should void within 24 hours after circumcision. Monitor for urinary retention. Educate the parents that the newborn needs to void within 24 hours after the circumcision.

F. METABOLIC AND GASTROINTESTINAL SYSTEM

— **Digestion:** Simple carbohydrates → digestible; fats → limited digestion due to lack of lipase.
— **Proteins:** Partially digested → may act as antigens → possible allergic reaction.
— **Stomach capacity:** <10 mL at birth → ~90 mL by Day 10.
— **Intestinal peristalsis:** Rapid; bowel emptying 2.5–3 hours.
— **Feeding:** Breast/chest-feeding can begin immediately after birth; initial formula ≤30 mL per PHCP/agency protocol.
— **Feeding reflexes:** Rooting → sucking → swallowing.
— **Feeding**
    - Breast/chest-feeding: every 2–3 hours
    - Formula: every 3–4 hours
    - Formula: minimum 30 mL (1 oz) by Day 3
    - Follow PHCP/agency protocol.
— **Burping**
    - Burp during and after feeding.
— **Assess feeding tolerance**
    - Observe for regurgitation or vomiting.
— **Position after feeding**
    - Position newborn on the right side after feeding.
    - Do not use side-lying for sleep because newborn may roll prone.

- Prone sleeping is contraindicated → increases SIDS risk.
- Turning the head to the right is another option.

— **Stool Assessment**
- **Meconium:** Greenish-black, thick, sticky, tarlike, usually passed within 24 hours, may be delayed in premature infants
- **Transitional stool:** Greenish-brown, looser than meconium
- **Human milk-fed:** Seedy, yellow stools,
- **Formula-fed:** Pale yellow to light brown stools

— **Newborn Screening**
- Perform newborn screening before discharge as prescribed.
- Includes testing for phenylketonuria (PKU).
- Ensure sufficient protein intake before screening.
- According to the source, newborn should receive **formula or human milk for 24 hours** before screening.

G. NEUROLOGICAL SYSTEM

1. Newborn head size is proportionally larger than that of an adult because of cephalocaudal development.
2. **Myelinization** of nerve fibers is incomplete, so primitive reflexes are present.
3. Fontanels are open to allow for brain growth.
4. Assess for abnormal head size and a bulging or depressed anterior fontanel.
5. Measure and graph the head circumference in relation to chest circumference and length.
6. Assess the newborn's **movements**, noting **symmetry**, **posture**, and **abnormal movements.**
7. Observe jitteriness, marked tremors, and seizures.
8. Test the **newborn's reflexes.**
9. Assess for lethargy.
10. Assess pitch of cry; a high-pitched cry may indicate a neurological or other disorder.

H. THERMAL REGULTORY SYSTEM

1. Prevent cold stress
2. Newborns do not shiver to produce heat.
3. Newborns have brown fat deposits, which produce heat.
4. Heat loss can occur via **evaporation, radiation, convection, conduction.**
5. Prevent heat loss resulting from evaporation by keeping the newborn dry and well wrapped with a blanket.
6. Prevent heat loss resulting from radiation by keeping the newborn away from cold objects and outside walls.
7. Prevent heat loss resulting from convection by shielding the newborn from drafts.
8. Prevent heat loss resulting from conduction by performing all treatments on a warm, padded surface.
9. Keep the room temperature warm.
10. Take the newborn's axillary temperature every 30 minutes for the first 2 hours of life until the newborn is stable, every 4 hours for the remainder of the first 24 hours, and then every shift (as per agency protocol).

I. REFLEXES

1. Sucking and Rooting

— *Stimulus:* Touch newborn's lip, cheek, or corner of the mouth with a nipple.
— *Response:* Turns head toward the nipple, opens mouth, and sucks
— *Disappearance Timeline:* Rooting disappears adter 3-4 months, may persist for 1 year.
— *Notes:* Important for feeding behavior.

2. Swallowing Reflex

— *Stimulus:* Occurs after sucking and obtaining fluids.
— *Response:* Swallowing without gagging, coughing, or vomiting.
— *Disappearance Timeline:* N/A
— *Notes:* Automatic response in coordination with sucking.

3. Tonic Neck (Fencing)

— *Stimulus:* Gently turn the head to one side while sleeping.
— *Response:* Arm and leg on the side the face is turned extend, opposite side flexes.
— *Disappearance Timeline:* Disappears withing 3-4 months
— *Notes:* Linked to hand-eye coordination

4. Palmar-Plantar Grasp

— *Stimulus:* Place fingers in palm and at the base of toes.
— *Response:* Fingers curl around the examiner's finger, toes curl downward
— *Disappearance Timeline:* Palmar disappears withing 3-4 months, while Plantar is within 8 months.
— *Notes:* Important for bonding and grasp reflex.

5. Moro Reflex

— *Stimulus:* Hold newborn in a semi-sitting position and let head/trunk fall back.
— *Response:* Sharp extension and abduction of arms, then flexion and adduction.
— *Disappearance Timeline:* Disappears by 6 months
— *Notes:* Delayed disappearance may indicate neurological issues.

6. Startle Reflex

— *Stimulus:* Make a loud noise/clap hands
— *Response:* Arms adduct, elbows flex, hands stay clenched
— *Disappearance Timeline:* Disappears within 4 months
— *Notes:* Similiar to Moro reflex but distinct in response.

7. Pull-to-Sit Response

— *Stimulus:* Pull newborn up by wrist while supine
— *Response:* Head lags, then momentarily levels with chest before falling forward.

— *Disappearance Timeline:* Dependent on muscle tone and maturity.
— *Notes:* Used to assess general muscle tone and maturity.

8. Babinski's Sign
— *Stimulus:* Stroke foot from heel upward along lateral side
— *Response:* Toes hyperextend, big toe dorsiflexes
— *Disappearance Timeline:* Disappears after 1 year
— *Notes:* Absence may indicate neurological issues.

9. Stepping or Walking Reflex
— *Stimulus:* Hold newborn upright, allow one foot to touch surface
— *Response:* Alternating stepping movements
— *Disappearance Timeline:* N/A
— *Notes:* Early indicator of motor development.

NEWBORN SAFETY

A. NEWBORN IDENTIFICATION

1. Information bracelets are applied to the birthing parent and newborn immediately after birth and before the birthing parent and newborn are separated; in addition, identification pictures of the newborn and footprints from the newborn may be obtained before the newborn leaves the birthing parent's side in the delivery room.
2. The bracelets include name, sex, date, time of birth, and identification numbers.
3. Some agencies use identification bracelets that have radiofrequency transmitters that set off alarms if the newborn is removed from a certain area.
4. Agencies also conduct unit and hospital-wide drills to prevent newborn abductions.

B. NEWBORN ABDUCTION

1. The birthing parent is taught to check the identification of any person who comes to remove the infant from the room and is taught other precautions to prevent newborn abduction (nurses must be wearing photo identification or some other security badge)
2. Closed-circuit televisions, code-alert bands, computer monitoring systems, or other monitoring systems may be used in some agencies.
3. The newborn is wheeled in a bassinet, not carried in a staff member's arms.

PARENT TEACHING

A. FORMULA FEEDING

1. Teach sterilization techniques if the water supply is located in areas where the purification process of the water is questionable.
2. Remind the parents not to heat the bottle of formula in a microwave oven.
3. Inform the parents that formula is a sufficient diet for the first 4 to 6 months.
4. Assess the parents ability to burp the newborn.

B. BREAST-FEEDING/CHEST FEEDING

1. Assess the newborn's ability to attach to the parent's breast/chest and suck
2. Teach the parent how to pump breasts/chest and how to store breasts/chest milk properly.
3. Inform the parent that human milk is a sufficient diet for the first 4 to 6 months.
4. Give the parents the phone numbers of local organizations that offer support to breast-feeding/ chest-feeding parents.

C. BATHING

1. Parents may choose to delay bathing the newborn for up to 8 hours after birth to help keep the skin conditioned.
2. Bathe the newborn in a warm room before feeding.
3. Have all equipment for bathing available.
4. Use a mild soap (not on the face).
5. Proceed from the cleanest area to the dirtiest.
6. Clean eyes from the inner canthus outward.
7. Special care should be taken to clean under the folds of the neck, underarms, groin, and genitals.
8. Make bath time enjoyable for the newborn and the parent.

D. UNCIRCUMSISED NEWBORN

1. Inform parents that the foreskin and glans are two similar layers of cells that separate from each other and that the separation process normally is complete by 3 to 6 years of age, although the layers can remain adhered until puberty.
2. Instruct parents not to pull back the foreskin, but to allow natural separation to occur; pulling back on the foreskin could be painful and could cause bleeding, adhesions, and infection.
3. Inform parents that as the process of separation occurs, sloughed cells build up between the layers of the foreskin and the glans, and that when retraction occurs, daily gentle washing of the glans with soap and water is sufficient to maintain adequate cleanliness.

E. STIMULATION

— Providing stimulation to the newborn such as touching, cuddling, or talking is an important intervention.

GROWTH AND DEVELOPMENT

A. GROWTH

— Quantitative/ measurable changes
— Parameters
- Weight
- Height

— Weight gain patterns
- 2× birth weight → 5–6 months
- 3× birth weight → 1 year
- 4× birth weight → 2–2½ years

— Height growth patterns
- 1 inch/month → 1–6 months
- 1.5 inches/month → 7–12 months
- 50% increase in birth length → first year

— Height comparison by age

B. DEVELOPMENT
— Qualitative changes
— Assessment:
- Observation
- Parent feedback
- Developmental screening test

— Developmental domains
- Language
- Personal/Social
- Fine Motor-Adaptive
- Gross Motor

C. DEVELOPMENTAL SCREENING
— Denver Developmental Screening Test (DDST)
— Metro Manila Developmental Screening Test (MMDST)

COGNITIVE DEVELOPMENT

A. DEFINITION
— Learning
— Thinking
— Understanding
— Problem-solving

B. INTELLIGENCE
— Mental age
— Chronological age
— IQ:
- Normal IQ: 90–110
- Gifted child: IQ ≥130

C. BASIC DIVISIONS OF LIFE
— Prenatal
— Infancy
- Neonatal
- Formal infancy

— Early childhood
- Toddler
- Preschool

— Middle childhood
- School age

— Late childhood
- Pre-adolescent
- Adolescent Pedia

PRINCIPLES OF GROWTH AND DEVELOPMENT

— **Continuous process** - "Womb to tomb"
— **Asynchronous growth** - Body parts do not grow at the same rate.
— **Each child is unique** - Individual differences in development.
— **Predictable patterns**
- Cephalocaudal - Head → toe
- Proximodistal - Center → extremities
- Symmetrical development
- Mass-specific / differentiation - Gross → fine/refined skills

FACTORS AFFECTING GROWTH AND DEVELOPMENT

A. HEREDITARY
— Race
— Sex
— Intelligence
— Nationality

B. ENVIRONMENT
— Nutrition
— Socioeconomic status
— Health
— Ordinal position/birth order
— Parent-child relationship

SIGMUND FREUD'S PSYCHOSEXUAL THEORY OF PERSONALITY DEVELOPMENT

Stage	Age Range	Site of Gratification	Main Activities / Focus	Key Task / Possible Outcome
Oral Phase	0–18 months	Mouth	Sucking, biting, chewing (e.g., breastfeeding, pacifier use)	Dependence on caregivers for comfort. Poor stimulation may lead to oral fixation (smoking, nail-biting, overeating).
Anal Phase	19 months–3 years	Anus	Pleasure from bowel/bladder control; potty training	Learning control. Too strict or lenient training may lead to traits like stubbornness, messiness, or excessive orderliness.
Phallic Phase	4–6 years	Genitals	Awareness of body and gender differences	Resolving feelings related to Oedipus/Electra complex; healthy understanding of body and identity.
Latent Phase	7–12 years	Dormant libido	Focus on school, friendships, hobbies, skills	Development of social and cognitive abilities; psychosexual energy is redirected to learning and social life.
Genital Phase	12–18 years	Genitals	Mature romantic and sexual interests; dating	Achieving sexual maturity, forming relationships, and balancing work, love, and social life.

ERIK ERIKSON'S PSYCHOSOCIAL THEORY

Stage	Age Range	Psychosocial Conflict	Main Focus / Activity	Outcome if Resolved
1. Trust vs. Mistrust	0–18 months	Dependence on caregivers	Receiving consistent care, feeding, comfort	Develops **trust, security, hope**
2. Autonomy vs. Shame & Doubt	18 months–3 years	Independence and self-control	Making choices, toilet training	Develops **will, confidence, independence**
3. Initiative vs. Guilt	4–6 years	Taking initiative in activities	Play, exploration, asking questions	Develops **purpose and initiative**
4. Industry vs. Inferiority	7–12 years	Mastery of skills	School work, projects, teamwork	Develops **competence and achievement**
5. Identity vs. Role Confusion	12–18 years	Developing sense of self	Exploring roles, values, interests	Develops **identity and fidelity**
6. Intimacy vs. Isolation	18–25/30 years	Forming close relationships	Romantic relationships, friendships	Develops **love and intimacy**
7. Generativity vs. Stagnation	30–45 years	Contributing to society	Parenting, mentoring, productive work	Develops **care and productivity**
8. Ego Integrity vs. Despair	45+ years	Reflection on life	Reviewing life achievements	Develops **wisdom and life satisfaction**

JEAN PIAGET'S STAGES OF COGNITIVE DEVELOPMENT

Stage	Age Range	Main Characteristics	Examples / Key Skills
Sensorimotor Stage	0–2 years	Learning through senses and actions; development of object permanence	Reflexes (grasping), repeating actions (thumb sucking), exploring objects, goal-directed behavior, symbolic play (pretending a block is a car)
Pre-operational Stage	2–7 years	Symbolic thinking, language development, egocentrism, animism	Believing the sun follows them, thinking toys have feelings, curiosity about cause and effect
Concrete Operational Stage	7–12 years	Logical thinking about concrete objects; understanding conservation and reversibility	Knowing water amount stays the same in different containers, solving puzzles, sorting by size/shape
Formal Operational Stage	12 years and above	Abstract thinking, hypothetical reasoning, future planning	Solving algebra problems, considering career choices, thinking about possibilities and consequences

KOHLBERG'S THEORY OF MORAL DEVELOPMENT

Level	Stage	Age Range	Main Idea	Example
Preconventional	**Stage 1: Punishment & Obedience**	0–3 years	Right and wrong based on avoiding punishment	Child avoids touching a stove because they might get punished or hurt
	Stage 2: Individualism & Exchange	4–7 years	Behavior motivated by rewards and self-interest	Child shares a toy expecting something in return
Conventional	**Stage 3: Interpersonal Relationships (Good Boy/Good Girl)**	4–10 years	Desire to gain approval and be seen as good	Child does homework to be praised by parents/teachers
	Stage 4: Maintaining Social Order	10–12 years	Following rules to maintain order in society	Child obeys school rules because rules keep things fair
Postconventional	**Stage 5: Social Contract & Individual Rights**	Older than 12	Laws are important but can be questioned if unjust	Teen questions unfair laws and values justice
	Stage 6: Universal Ethical Principles	Adulthood (rare)	Moral decisions based on universal principles like justice and human rights	Individuals stand against injustice even if it breaks laws

MIDTERMS: CMPA411 (COMPETENCY APPRAISAL 1) - MODULE #9: PEDIATRIC NURSING CARE OF AT RISK

UNIVERSAL PEDIATRIC NURSING FRAMEWORK

1. **Airway and Breathing**: Is the airway open? Is the child breathing effectively? Look for apnea, grunting, chest retractions, cyanosis, stridor, or a silent chest.
2. **Circulation and Perfusion:** Check heart rate, pulses, capillary refill, skim temperature, blood pressure, mental status and urine output. Hypotension can be late sign of pediatric shock.
3. **Neurologic Status:** Assess responsiveness, tone, seizures, pupils when indicated, and age-appropriate behavior.
4. **Hydration and Glucose:** consider dehydration, hyperglycemia, hyperglycemia, and electrolyte abnormalities in a child who is lethargic, vomiting, or feeding poorly
5. **Age and Weight:** Obtain an accurate weight in kilogramms, assess developmental stage, and calculate medication doses carefully
6. **Family-centered care**: Explain procedures in developmentally appropriate language, involve caregivers when safe, and protect the child's privacy and dignity.

HYPOTENSION

— Hypotension can be a late sign of pediatric shock. Do not wait for a low blood pressure before recognizing poor perfusion. A child can be in shock with a normal blood pressure. Recognize poor perfusion early; do not wait for hypotension.
— Early intervention can prevent progression to cardiovascular collapse.

The Early Warning Signs of Shock:

— **T** — Tachycardia
— **C** – Capillary refill prolonged
— **C** – Cool or mottled extremities
— **M** - Mental status changes
— **U** - Urine output decreases

These findings must be interpreted together and in the context of the child's age and condition. No single sign is sufficient to diagnose shock.

— Do not interpret ABCDE as a rigid checklist that delays resuscitation. If a child is unresponsive and not breathing normally, choose the action that addresses the most immediate threat to life.
— Stabilization comes before routine teaching, documentation, or nonurgent investigations. Activate emergency support according to the clinical setting.

PART 1: NEWBORN AND INFANT

A. PREMATURITY, POST-TERM BIRTH, SGA AND LGA

Term	Definition	Major Risks
Pre-term	Born before 37 completed weeks	Respiratory distress, apnea, hypothermia, hypoglycemia, infection, feeding difficulty.
Post-term	Born at por after 42 completed weeks	Meconium aspiration, birth injury, placental insufficiency
SGA	Birth weight below the 10th percentile for gestational age.	Hypoglycemia, hypothermia, polycythemia
LGA	Birth weight above the 90th percentile for gestational age.	Birth trauma, hypoglycemia, respiratory complications

— ***Remember:*** SGA does not automatically mean premature. A term infant may be SGA and a preterm infant may be appropriate for gestational age.

1. PREMATURE INFANT

— A preterm infant is born before 37 weeks of completed gestation.

Pathophysiology:

— Immature organs have difficulty maintaining normal body functions.
— Immature lungs may lack adequate surfactant
— Immature skin loses heat and water readily
— Immature feeding and swallowing coordination makes oral feeding difficult.

Clinical Manifestations:

— Thin/skin and little subcutaneous fat.
— Weak cry, poor suck and swallow coordination.
— Respiratory distress, apnea or oxygen desaturation
— Temperature instability.
— Feeding intolerance, poor weight gain and increased infection risk.

Priority Nursing Interventions:

1. Maintain a neutral thermal environment, monitor temperature closely.
2. Assess respiratory effort, oxygenation and apnea.
3. Monitor blood glucose, feeding tolerance, weight and fluid balance.
4. Use strict hand hygiene and infection-provention measures.
5. Provide expressed breast milk and an appropriate feeding method based on maturity and clinical stability.
6. Support kangaroo mother care when clinically appropriate. WHO guidance emphasizes thermal care, feeding support and kangaroo mother care for preterm and low-birth-weight infants.

Question: A premature newborn has grunting. retractions and oxygen desaturation. Should the nurse begin with feeding?
Answer: Do not feed an infant who is in significant respiratory distress or cannot coordinate sucking swallowing and breathing safely stabilize the infant and follow the prescribed feeding plan.

2. POST-TERM INFANT

— A post-term infant is born at or beyond 42 completed weeks.
— Placental function may decline in some post-term pregnancies.
— Meconium passage before birth may lead to aspiration, while reduced placental reserve can contribute to fetal compromise.

Priority Nursing Interventions:
— Respiratory assessment
— Temperature
— Glucose monitoring when indicated
— Assessment for birthptrauma
— Early recognition of meconium aspiration or other respiratory complications.

3. SMALL FOR GESTATIONAL AGE (SGA)

— SGA means the infant's birth weight is below the 10th percentile for gestational age. It is a description of size, not a synonym for prematurity
— These infants may have limited energy reserves and increased risk of hypoglycemia and hypothermia.

Priority Nursing Interventions:
— Monitor glucose according to newborn risk protocols.
— Maintain warmth.
— Assess feeding effectiveness and weight.
— Observe for jitteriness, lethargy, apnea or seizures, which can indicate significant hypoglycemia or another neurologic problem.

4. LARGE FOR GESTATIONAL AGE (LGA)

— LGA means birth weight is above the 90th percentile for gestational age. Maternal diabetes is an important risk facto but not the only cause.
— An infant exposed to high maternal glucose may produce excess insulin. After birth, the maternal glucose supply stops while insulin may remain elevated, causing hypoglycemia).
— SGA, LGA and infants of diabetic mothers may all require glucose monitoring. Never assume that a large newborn cannot become hypoglycemic.

B. NEONATAL HYPERBILIRUBINEMIA

Assessment findings:
— Yellow discoloration of the skin and sclera.
— Poor feeding or excessive sleepiness.
— Jaundice begins during the first 24 hours, which is a warning sign.
— Dark urine or pale stools, which require evaluation for possible conjugated hyperbilirubinemia.
— Abnormal tone, a high-pitched cry or aching in severe bilirubin neurotoxicity.

Pathophysiology:
— Bilirubin is produced from red blood cell breakdown.
— The newborn liver has limited capacity to conjugate and eliminate bilirubin.
— Very high unconjugated bilirubin can enter the brain and cause **acute bilirubin encephalopathy or kernicterus**.

Priority Nursing Interventions:
1. Assess the timing of jaundice and relevant risk factors.
2. Measure bilirubin using the recommended method and interpret it against the infant's age in hours, gestational age and neurotoxicity risk factors.
3. Support adequate breastfeeding and assess milk transfer, weight loss and hydration.
4. Administer phototherapy when ordered and indicated by the applicable treatment guideline.
5. During phototherapy, protect the eyes according to the device protocol, expose appropriate skin, monitor temperature and hydration, and assess bilirubin response.
6. Escalate urgently for signs of acute bilirubin encephalopathy.

Very high unconjugated bilirubin can cause acute bilirubin encephalopathy and permanent
— Do not treat jaundice in the first 24 hours as automatically physiologic.
— Do not recommend direct sunlight as a substitute for prescribed phototherapy.
— Do not use a single bilirubin cutoff for every newborn.

C. MECONIUM ASPIRATION SYNDROME

— Meconium-stained amniotic fluid does not automatically mean the infant has aspirated meconium.

DIFFERENCE BETWEEN MECONIUM STAINED AND MECONIUM ASPIRATION

Meconium-stained Fluid	Meconium Aspiration Syndrome
— Meconium is present in the amniotic fluid. — Does not necessarily mean the baby has a lung problem	— Meconium enters the baby's airway and causes lung problems. — Can cause respiratory distress and low oxygen levels. — Manifestations: Respiratory distress. grunting, retractions, cyanosis and oxygen desaturation.

— Meconium in the fluid does not equal meconium in the lungs.
— If meconium enters the lungs, remember:

- Blockage → Inflammation → Surfactant disruption → Poor oxygenation
- That is why a newborn with meconium-stained fluid needs appropriate assessment after birth, but the presence of stained fluid alone does not establish meconium aspiration syndrome.

Priority Nursing Interventions:

1. Assess airway and breathing immediately.
2. Support oxygenation and ventilation according to neonatal resuscitation guidance.
3. Monitor for persistent pulmonary hypertension and air leaks.
4. Maintain warmth and monitor glucose as indicated.
5. Follow neonatal resuscitation guidance: routine tracheal suctioning is not recommended solely because meconium-stained fluid is present. Resuscitation focuses on effective ventilation when indicated.
6. Assess first. Ventilate when indicated, suction for obstruction-not just for meconium.

The priority is effective ventilation-not routine suctioning of every meconium-exposed newborn.

D. FETAL ALCOHOL SPECTRUM DISORDERS

— Prenatal alcohol exposure may affect fetal growth, brain development, behavior and learning. Fetal alcohol syndrome is one diagnosis within the broader fetal alcohol spectrum.

— Possible findings include growth restriction, characteristic facial features, developmental delay, difficulties with attention and impulse control, and learning problems.

Priority Nursing Interventions:

1. Document developmental and behavioral findings objectively like difficulty coordinating sucking. swallowing, and breathing during feeding.
2. Support early developmental assessment and referral
3. Use predictable routines and simple, clear instructions
4. Educate caregivers without blame or stigma
5. Reinforce that avoiding alcohol during pregnancy is the safest prevention approach.

There is no treatment that reverses prenatal alcohol-related brain injury: care focuses on early recognition, developmental support and managing associated difficulties.

E. NEONATAL SEPSIS — A LIFE-THREATENING

Priority:

— Treat the sick newborn in an emergency.

— Neonatal sepsis can progress rapidly, and signs may be subtle.

Risk Factors:

— Prematurity and low birth weight

— Maternal infection.

— Prolonged rupture of membranes in relevant circumstances.

— Invasive procedures and prolonged hospitalization.

Pathophysiology:

— Infection can trigger a systemic inflammatory response, causing poor tissue perfusion. respiratory dysfunction, metabolic abnormalities and, in severe cases, shock and organ failure

Findings:

Early or Possible Findings	Finding Suggesting Deterioration
— Poor feeding, lethargy, irritability — Temperature instability — Weak cry, reduced activity	— Apnea or gasping — Respiratory distress or cyanosis — Poor perfusion, prolonged capillary refill
— Abdominal distension or vomiting	— Hypotension, seizures or reduced urine output

— These findings require urgent assessment because they may occur in sepsis and other serious newborn conditions.

Priority Nursing Interventions:

— Recognize and escalate immediately.

— Support airway, breathing and circulation.

— Monitor temperature, glucose, oxygenation, perfusion and urine output.

— Obtain ordered cultures and investigations promptly, without delaying urgent treatment.

— Administer prescribed antimicrobials on time.

— Maintain warmth and aseptic technique.

CONGENITAL AND DEVELOPMENTAL ANOMALIES

A. AUTISM SPECTRUM DISORDER (ASD)

— ASD is a neurodevelopmental condition, not a structural congenital anomaly. It is included here because it affects pediatric nursing assessment and long-term-care.

— Possible features include differences in social communication, restricted or repetitive behaviors, strong preferences for routines, and sensory sensitivities. Presentation varies widely.

Priority Nursing Interventions:

— Use developmentally appropriate language and concrete instructions.

— Reduce unnecessary sensory stimulation

— Explain procedures before performing them.

— Allow additional processing time and involve caregivers in identifying effective communication strategies.

— Avoid assuming that a child who speaks fluently has no communication or sensory support needs.

— Refer for developmental assessment when concerns are identified.

The nurse supports the child's communication, safety and participation. ASD is not caused by poor parenting.

B. VENTRICULAR SEPTAL DEFECT (VSD)

— A VSD is an opening between the ventricles. Blood commonly flows from the higher-pressure left ventricle to the right ventricle, increasing pulmonary blood flow. A large defect can lead to heart failure and poor growth.

Manifestations:
— A characteristic murmur
— Tachypnea and difficulty feeding
— Sweating during feeding
— Poor weight gain
— Signs of heart failure in larger defects

Nursing Management:
— Assess respiratory effort, feeding tolerance and growth.
— Provide small, frequent, energy-dense feeds as prescribed.
— Cluster care to conserve energy.
— Administer prescribed medications and monitor response.
— Teach caregivers to report worsening breathing, sweating, feeding intolerance or reduced urine output

A small VSD may cause few symptoms; a large VSD may cause heart failure. The size and hemodynamic effect matter.

C. TETRALOGY OF FALLOT (TOF)

— TOF includes four abnormalities:
1. Ventricular Septal Defect.
2. Pulmonary Stenosis or Right Stenotic Ventricular Outflow Obstruction
3. Overriding aorta.
4. Right Ventricular Hypertrophy.

— The degree of obstruction influences the severity of cyanosis.

Tet spell: A sudden episode of increased cyanosis and respiratory distress, often during crying or feeding.

Priority actions during a suspected tet spell:
— Call for urgent assistance.
— Place the infant or child in a knee-chest position if appropriate and tolerated; this can increase systemic vascular resistance and reduce right-to-left shunting.
— Calm the child and minimize distress.
— Administer oxygen and medications or fluids as prescribed under emergency protocols.
— Monitor oxygenation, perfusion, and responsiveness.
— Knee-chest positioning is associated with a hypercyanotic spell in TOF, not the routine treatment for every cyanotic child.

D. CLUBFOOT

— Clubfoot, or congenital talipes equinovatis, involves an inward and downward position of the foot.

Nursing Management:
— Assess foot position, skin integrity, circulation and movement.
— Explain the usual serial casting and bracing approach.
— Monitor cast edges, swelling, color, warmth and toe movement.
— Teach caregivers to keep the cast dry and seek help for cold, blue, swollen toes or signs of impaired circulation
— Reinforce adherence to bracing after correction to reduce recurrence.
— Reassess distal circulation and neurovascular status after casting.

Types of Talipes (Clubfoot)



Treatment:
Ponseti Method
— a nonsurgical treatment
— gradually corrects the infant's foot position
— 3 methods: manipulation-casting-bracing

Nursing Management:
— Check the toes for color, warmth, movement, swelling, and circulation after casting.
— Teach caregivers to keep the cast clean and dry.
— Report blue or pale toes, cold toes, increasing swelling, unusual crying, or a cast that becomes too tight or slips.
— Never insert objects inside the cast to scratch the skin.
— Emphasize brace adherence because failure to use the brace as prescribed increases the risk of clubfoot recurrence.

The Ponseti method is the preferred initial treatment for most infants with idiopathic congenital clubfoot. Surgery is not usually the first-line treatment.

D. CLEFT LIP AND CLEFT PALATE

— **Cleft lip:** Failure of the maxillary prominence to fuse with the medial nasal prominence.
— **Cleft palate:** Failure of the palatal shelves to elevate, meet, and fuse with each other and related structures.
— These conditions can interfere with feeding, speech, hearing and dental development. Cleft palate particularly impairs the infant's ability to create the suction needed for effective feeding.

Risk Factors
1. Genetics
2. Intrapartum infections
3. Tobacco, drugs, alcohol use during pregnancy
4. Vitamin Deficiencies (Vitamin B9 and B6)

Treatment
— Surgery is recommended to be done within the first 18 months

Preoperative Nursing Care
— Assess feeding effectiveness, weight gain and hydration.
— Position and feed according to the individualized plan, often upright or semi-upright.
— Use a specialized feeder (1 ampule nipple or wide nipple) when indicated.
— For unilateral cleft lip, feed on unaffected side
— Allow frequent pauses and swallow time and burping; monitor for coughing, choking or respiratory distress.
— Parent education is KEY to help the baby feed effectively and for parents to feel supported.

Postoperative Nursing Care
— Follow the surgeon's feeding and wound-care instructions.
— Maintain airway patency and observe for bleeding
— Prevent injury to the surgical site using the prescribed approach; avoid the use of pacifier because sucking can disrupt the suture and use of elbow immobilizer to avoid the baby on touching the surgical site
— Monitor pain, hydration and nutritional intake.

Cleft lip repair focuses on protecting the suture line, while cleft palate repair focuses on protecting the surgical repair inside the mouth. Cleft palate increases feeding difficulty and the risk of nasal regurgitation. Feeding support is a central nursing priority.

F. IMPERFORATE ANUS

— This is an anorectal malformation in which the anal opening is absent, abnormally positioned or not connected normally to the rectum.
— Clues: No visible anal opening, failure to pass meconium, abdominal distension or abnormal passage of stool through the urinary tract or genital tract.

Priority Nursing Interventions
— Assess the perineum and passage of meconium.
— Keep the infant NPO when obstruction or surgery is being evaluated.
— Notify the neonatal or surgical team promptly
— Monitor abdominal distension, vomiting and signs of obstruction.
— Do not insert a rectal thermometer, enema or rectal instrument without a specific clinical order and appropriate specialist direction.

G. HISCHSPRUNG DISEASE

— A segment of bowel lacks ganglion cells, preventing normal relaxation and peristalsis. This creates functional obstruction.

Classic clues:
— Failure to pass meconium within the expected period
— Abdominal distension
— Bilious vomiting
— Chronic constipation in older infants or children.
— Explosive stools after rectal examination may occur, but examination should be performed only by an appropriately trained clinician.
— *Major complication:* Hirschsprung-associated enterocolitis, which may cause fever, abdominal distension, diarrhea, lethargy and rapid deterioration.

Priority Nursing Interventions
— Recognize obstruction and enterocolitis garly
— Keep NPO when indicated and notify the team
— Monitor abdominal girth, emesis, stool, hydration and perfusion
— Prepare for diagnostic and surgical management.
— Follow the prescribed bowel-management plan, do not independently perform rectal procedures.

Bilious vomiting in a newborn or infant is a surgical warning sign until evaluated.

H. HYDROCEPHALUS

— Hydrocephalus involves abnormal accumulation of cerebrospinal fluid and enlargement of the cerebral ventricles, sometimes with increased intracranial pressure.
— The main problem is an imbalance between CSF production circulation, and absorption, resulting in fluid buildup that can compress or damage brain tissue.

Infant Findings:
— Rapidly increasing head circumference.
— Bulging or tense fontanelle when calm and upright.
— Widely separated cranial sutures.
— Poor feeding, vomiting, irritability or lethargy.
— Downward deviation of the eyes, sometimes described as "sunsetting."
— Older-child findings may include headache, vomiting. visual changes, gait changes and altered mental status.

Priority Nursing Interventions

— Measure and trend head circumference accurately.
— Assess neurologic status and signs of increased intracranial pressure.
— Monitor feeding and vomiting.
— After shunt placement, observe for infection or shunt malfunction.
— *Shunt malfunction or infection clues:* New Vomiting, letargy, irrilaojlity. fever, altered consciousness, headache or increasing head size in infants require prompt evaluation.

PART 2: TODDLER

A. POISONING

— Toddlers explore by touching, tasting and mouthing objects.
— Common hazards include medicines, cleaning products, pesticides, hydrocarbons, button batteries and magnets.
— TAKE NOTE!!! Unknown substance + toddler + symptoms = assess first. do not induce vomiting.

Initial Nursing Assessment

— Determine the child's airway, breathing, circulation, mental state and vital signs.
— Identify the substance, amount, route and approximate time of exposure.
— Obtain the container, label or a photograph if safely availed
— Contact the local poison-information service or emergency team and follow their instructions.
— Monitor for seizures, respiratory depression, burns, dysrhythmias oF altered consciousness.

What should the nurse avoid?

— Do not induce vomiting.
— Do not give milk, food, activated charcoal or home remedies unless specifically directed by an appropriate poison specialist or treating clinician.
— Do not delay emergency care while trying to identify every detail.
— Do not give anything orally to a child with impaired consciousness or swallowing.

SPECIAL EMERGENCIES

— **Button battery ingestion:** Urgent emergency assessment is required because serious esophageal injury can develop quickly.
— **Caustic ingestion:** Do not induce vomiting or attempt chemical neutralization.
— **Hydrocarbon Ingestion:** Vomiting increases the risk of aspiration.
— **Opioid exposure with respiratory depression:** Support breathing and activate emergency response; naloxone may be indicated under protocol.

Store medicines and chemicals in original containers, locked and out of reach. Never call medicine "candy." Use child-resistant packaging correctly, while remembering that it is not childproof.

B. CEREBRAL PALSY (CP)

— Cerebral palsy (motor impairment) describes a group of permanent disorders of movement and posture caused by a nonprogressive disturbance in the developing fetal or infant brain. Functional difficulties may change as the child grows.

Manifestations:

— Delayed motor milestones.
— Abnormal muscle tone, such as spasticity or hypotonia.
— Persistent primitive reflexes or abnormal posturing.
— Scissoring of the legs or toe-walking in some children
— Feeding, swallowing, communication or mobility difficuties.
— Possible seizures, vision impairment or learning difficulties.

Priority Nursing Interventions:

1. Assess mobility, tone, pain, nutrition and swallowing safety.
2. Use aspiration precautions and a prescribed feeding plan where indicated.
3. Encourage positioning, range-of-motion and physical, occupational and speech therapy.
4. Monitor skin integrity, contractures and equipment fit.
5. Support caregivers and promote inclusion and independence

— Cerebral palsy is non-progressive in its underlying brain injury, but secondary complications and functional needs may change over time.

C. CHILD ABUSE AND NEGLECT

Forms:

— Physical abuse
— Sexual abuse
— Emotional abuse
— Neglect.

Nurses must recognize possible indicators while avoiding unsupported conclusions.

Warning Signs

— Injuries inconsistent with the explanation or the child's developmentel abilies
— Repeated injuries or delayed presentation for care.
— Bruises in unusual locations, patterned injuries or multiple injuries at different healing stages.
— A child who appears fearful of a caregiver
— Poor hygiene, malnutrition or unmet medical needs in a context suggesting neglect.
— Sexualized behavior or disclosures that raise concern.
— No single sign proves abuse. Accidental injuries can also occur, so the assessment must be careful and objective.

Nursing Responsibilities:

— Stabilize injuries and address immediate safety.

— Speak with the child calmly, using open ended, nonleading question
— Record the child's words as accurately as possible and document objective findings.
— Follow institutional safeguarding procedures and applicable Philippines reporting requirements.
— Refer to the appropriate child-protection team or authorities as required
— Do not promise secrecy, conduct an unauthorized festigation or confront suspected perpetrator in a way that increases danger.
— Protect the child and provide appropriate care.
— Documentation and reporting should follow the applicable protocol.

What should the nurse avoid?
— Do not write judgmental statements such as "The parent is abusive."
— Do not alter, exaggerate, or speculate about findings.
— Do not repeatedly question the child or conduct an independent investigation.
— Do not promise the child absolute secrecy; explain that information may need to be shared to help keep them
—Do not delay urgent care or required reporting while waiting for proof.
— Suspect abuse → Ensure safety → Document objectively → Report promptly → Maintain confidentiality.
— The nurse does not need to prove abuse before raising a concern through the appropriate safeguarding and reporting channels.

PART 3: PRE-SCHOOLER

A. ACUTE LYMPHOBLASTIC LEUKEMIA (ALL)

— ALL is a malignancy of immature lymphoid cells in the bone marrow.
— Abnormal cells interfere with normal production of red blood cells, platelets and functioning white blood cells.

THREE-MARROW FAILURE PATTERNS
— Occur when abnormal immature lymphoid cells multiply in the bone marrow and interfere with normal blood cell production.

Blood Cell Affected	Result	Sign and Symptoms
RBCs	Anemia	Pallor, fatigue, weakness, dizziness, SOB.
Platelets	Thrombocytopenia	Easy bruising, petechiae, gum bleeding, nosebleeds, prolonged bleeding.
WBCs	Reduced effective immune defense	Fever, recurrent infections, poor wound healing.

1. Anemia — decreased RBC production
— Leukemic cells crowd the bone marrow, reducing red blood cell production. The child receives less oxygen to tissues.
— ***Nursing priority:*** Assess pallor, fatigue, heart rate, respiratory status and activity tolerance.

2. Thrombocytopenia - decreased platelet production
— The marrow produces fewer platelets, increasing the fisk of bleeding.
— ***Nursing priority:*** Assess for petechiae, bruising, mucosal bleeding and sign of internal bleeding. Avoid unnecessary invasive procedures and administer prescribed treatment.

3. Impaired immune defense - reduced effective WBC function
— Normal white blood cell production may be suppressed. Even If the total WBC count is high, cells may be immature leukemic cells that cannot adequately fight infection.
— ***Nursing Priority:*** Monitor for infection and treat fever as urgent, especially during chemotherapy or suspected neutropenia, following the oncology protocol.

A child with ALL may have a high WBC count but still be vulnerable to serious infection. Always consider the quality and maturity of the white blood cells. not just the total count.

Other Manifestations:
— Bone or joint pain
— Lymphadenopathy
— Hepatosplenomegaly
— Reduced activity
These findings are not specific to leukemia, diagnostic evaluation is essential.

Priority Nursing Interventions:
— Monitor temperature, bleeding, pain, fatigue and nutritional status.
— Treat fever during chemotherapy or suspected neutropenia as urgent and follow the oncology fever protocol.
— Use infection-prevention measures.
— Monitor CBC and other tests as ordered.
— Administer chemotherapy and supportive medications safely
— Watch for tumor lysis syndrome during treatment: electrolyte disturbances, hyperuricemia and acute kidney injury may occur.
— Provide developmentally appropriate explanations and emotional support.

Do not assume that a child with leukemia is protected from infection simply because the total white blood cell count is high. The cells may be immature or dysfunctional.

B. HEMOPHILIA

— Hemophilia is an inherited bleeding disorder, most commonly involving deficiency of factor VIII (hemophilia A) or factor IX (hemophilia B).

Clinical Manifestations

— Prolonged bleeding.
— Deep muscle hematomas.
— Painful swollen joints from hemarthrosis.
— Excessive bleeding after process trauma.
— Serious intracranial bleeding at injury.

Priority Nursing Interventions:

— Assess bleeding location, severity, circulation and neurologic status.
— Administer prescribed clotting-factor replacement promptly when indicated.
— Apply appropriate pressure to external bleeding,
— Avoid unnecessary intramuscular injections and invasive procedures
— Avoid aspirin and other medications that increase bleeding risk unless specifically directed by the treating specialist.
— Encourage protective activities and individualized joint protection
— Teach caregivers when to seek emergency help.
— Emergency Clue: A child with hemophilia who sustains a head injury needs urgent assessment even if initially appearing well.

C. WILMS TUMOR

— Wilms tumor, or nephroblastoma, is a malignant kidney tumor that typically occurs in young children.

Classic Presentation:

— A painless abdominal mass, sometimes accompanied by abdominal pain, hematuria,
— Hypertension or fever.

Most Important Nursing Precaution:

— Do not repeatedly palpate the abdominal mass. Excessive manipulation may increase the risk of tumor rupture or spillage.

Other Nursing Priorities:

— Measure blood pressure.
— Assess pain, urine findings and general condition.
— Prepare the child and family for imaging, surgery and other prescribed treatment.
— Monitor for postoperative complications ard treatment-related effects.
— Provide emotional support and age-appropriate preparation.

D. ASTHMA

— Asthma is a chronic inflammatory airway disorder associated with variable airflow obstruction and bronchial hyperresponsiveness.
— During an exacerbation, airway inflammation, bronchoconstriction and mucus production narrow the airways.

Pathophysiology:

— Trigger → bronchial inflammation → bronchoconstriction → mucosal edema → increased mucus → airway narrowing → difficulty breathing.

Danger Signs

— Silent or nearly silent chest despite severe distress.
— Cyanosis, exhaustion or reduced consciousness
— Poor air entry or inability to speak in phrases.
— Worsening respiratory effort despite treatment.
— **Status asthmaticus** is a severe, prolonged asthma attack that does not respond adequately to initial standard treatment and can progress to life-threatening respiratory failure. It is a medical emergency, especially in children.

Common Medications:

1. Short-acting bronchodilator
— Example: Albuterol
— Purpose: Bronchodilation.
2. Corticosteroids
— Purpose: Reduce airway inflammation.

Priority Interventions

1. Assess severity and initiate emergency response when indicated.
2. Position the child upright if tolerated.
3. Administer prescribed rapid-acting inhaled bronchodilator and oxygen when indicated.
4. Administer systemic corticosteroids for exacerbations when prescribed.
5. Reassess respiratory effort, air entry, oxygenation and treatment response.
6. Prepare for escalation of respiratory support if the child deteriorates.

Long-Term Education:

— Demonstrate inhaler and spacer technique.
— Explain controller versus reliever medicines.
— Reinforce adherence to the individualized asthma action plan.
— Reduce tobacco-smoke exposure and identify personal triggers.

E. TYPE 1 DIABETES MELLITUS AND DIABETIC KETOACIDOSIS

— The main difference is that Diabetes Mellitus (DM) is a chronic disease, while Diabetic Ketoacidosis (DKA) is an acute, life-threatening complication that can develop in a person with diabetes.

1. **Type 1 Diabetes** results from autoimmune destruction of pancreatic beta cells, causing absolute insulin deficiency.

Classic symptoms
— Polyuria.
— Polydipsia.
— Polyphagia.
— Weight loss.

2. A child may present for the first time with **DKA**, a life-threatening emergency involving hyperglycemia, ketosis and metabolic acidosis.

DKA Warning Signs
— Vomiting and abdominal pain.
— Dehydration and tachycardia.
— Deep, labored Kussmaul respirations.
— Fruity-smelling breath.
— Lethargy or altered mental status.

Nursing Priorities
1. Assess airway, breathing, circulation and neurologic status.
2. Monitor glucose, electrolytes, ketones, acid-base status and fluid balance as ordered.
3. Administer isotonic IV fluid and insulin according to the pediatric DKA protocol.
4. Monitor potassium closely; insulin shifts potassium into cells, so serum potassium may fall during treatment.
5. Observe for cerebral injury, including worsening headache, altered consciousness, bradycardia or other neurologic changes.
6. Do not administer an insulin bolus routinely in pediatric DKA; follow the established pediatric protocol.

DKA treatment requires controlled correction and close monitoring. Do not simply give insulin without assessing potassium and the child's overall clinical condition.

Long-Term Management
— Blood glucose monitoring and individualized insulin therapy.
— Carbohydrate education and meal planning.
— Recognition and treatment of hypoglycemia.
— Sick-day management and ketone monitoring when indicated.
— School support, physical activity planning and psychosocial care.
— Hypoglycemia: Sweating, tremor. hunger, irritability. confusion or seizures may occur. If the child is conscious and can swallow safely, provide fast-acting carbohydrate according to the prescribed plan. If consciousness is impaired, do not give anything orally: follow emergency treatment protocol.

F. ACUTE RHEUMATIC FEVER

— It is an inflammatory complication that can follow infection with group A streptococci. It can affect the heart, joints, skin and central nervous system.
— Major clinical features traditionally taught using the Jones Criteria:
- Carditis.
- Migratory polyarthritis.
- Sydenham chorea.
- Erythema marginatum.
- Subcutaneous nodules

— Diagnosis depends on the appropriate combination of clinical criteria and evidence of preceding streptococcal infection, with exceptions addressed in clinical guidance.
— Carditis is especially important because it can lead to rheumatic heart disease and chronic valve damage.

Nursing Management:
— Assess heart rate, respiratory effort, chest discomfort, murmurs, edema and exercise tolerance.
— Administer prescribed antibiotics (Penicillin V or Benzathine Penicillin G and anti-inflammatory treatment)
— Monitor for heart failure when carditis is present.
— Encourage rest during active illness as prescribed.
— Teach the importance of secondary antibiotic prophylaxis when indicated to prevent recurrence.
— Migratory joint pain after streptococcal infection plus evidence of carditis should raise concern for acute rheumatic fever.

G. DENGUE FEVER AND SEVERE DENGUE

— Dengue illness may begin with fever, headache, retro-orbital pair, myalgia, nausea and rash. Some children develop severe dengue characterized by plasma leakage, shock, severe bleeding or organ impairment.
— The Critical Concept: Defervescence does not always mean recovery.
— The danger period can begin as the fever subsides.
— You may reassure families solely because the temperature has returned to normal.

WARNING SIGNS
— Severe or persistent abdominal pain or tend
— Persistent vomiting
— Mucosal bleeding
— Lethargy or restlessness
— Clinical fluid accumulation
— Liver enlargement
— Rising hematocrit with a rapidly falling platelet count in the appropriate clinical context.
— These warning signs require prompt clinical assessment.

Nursing Assessment
— Trend vital signs, mental status and peripheral perfusion
— Monitor intake and output, urine output and hydration.
— Review CBC, hematocrit and platelet trends as ordered.

— Assess for bleeding, abdominal pain, persistent vomiting and respiratory difficulty.
— Watch for narrowing pulse pressure, cold extremities, prolonged capillary refill or hypotension.

Nursing Management:
— Give oral fluids when appropriate and tolerated.
— Administer prescribed isotonic IV fluids for indicated cases, with careful reassessment.
— Monitor the child's response to every fluid intervention
— Avoid fluid overload, particularly as capillary leakage resolves.
— Use paracetamol/acetaminophen when indicated for fever or pain.
— Avoid aspirin and NSAIDs such as ibuprofen because they can increase bleeding risk.
— Avoid unnecessary intramuscular injections and invasive procedures in patients at bleeding risk.
— Follow current DOH and facility dengue protocols.
— A low platelet count alone does not fully determine severity.
— Do not assume the child is safe because fever has subsided.
— Do not give aspirin or ibuprofen for suspected dengue.
— Do not treat every child with dengue using the same IV fluid rate; management depends on clinical status and reassessment.

PART 4: ADOLESCENT

A. SEXUALLY TRANSMITTED INFECTION

Common STIs include:
— Chlamydia Gonorrhea
— Syphilis
— Trichomoniasis
— Genital herpes
— HIV and HPV
— STD

Some infections are asymptomatic, and symptoms alone cannot reliably identify the pathogen.

Nursing Assessment and Communication
— Use respectful, nonjudgmental language.
— Provide a private interview when appropriate.
— Explain confidentiality and its limits before sensitive questions.
— Assess symptoms, exposure, pregnancy possibility. coercion and safety
— Obtain consent and follow local laws and clinical protocols for testing, treatment and reporting.
— Educate about correct condom use, testing, partner evaluation and avoiding reinfection.

Important Symptoms That Need Assessment
— Genital discharge
— Dysuria
— Genital sores or ulcers
— Pelvic or lower abdominal pain
— Testicular pain
— Abnormal bleeding
— A known exposure

Clinical Priority:
— Severe pelvic pain
— Fever
— Pregnancy with concerning symptoms
— Testicular pain with sudden onset = Sexual assault requires prompt assessment.

COMMON STDs:

Infection	Description
Chlamydia	Often asymptomatic; can cause urethritis, cervicitis and pelvic inflammatory disease
Gonorrhea	Can cause urethritis, cervicitis and pelvic inflammatory disease; resistance affects treatment.
Syphilis	Classically begins with a painless chancre: later stages can involve multiple organ systems
Genital Herpes	Painful vesicles or ulcers may recur: antiviral treatment helps control symptoms.
HIV	Attacks immune function; testing and effective antiretroviral therapy are essential
HPV	Some types cause genital warts; oncogenic types can cause cervical and other cancers

B. HPV AND CANCER PREVENTION

— HPV is a group of viruses. Certain high-risk types are associated with cervical cancer and other anogenital and oropharyngeal cancers. Other types pause genital warts.
— The CDC recommends routine adolescent FPV vaccination and catch-up vaccination for eligible people who were not vaccinated earlier. The Philippine school-based immunization program targets Grade 4 female learners for HPV vaccination. The HPV vaccine helps prevent HPV infections that can cause cervical cancer and other HPV-related diseases– not treatment for an existing cancer.

High-Yield Points
— HPV is common and often asymptomatic.
— HPV infection does not automatically mean cancer will develop.
— HPV vaccination prevents infection from important cancer-causing HPV types.
— Vaccination works best when given before exposure.
— Vaccination does not treat an established HPV infection and does not eliminate the need for age-appropriate cervical cancer screening later in life.

Nursing Responsibilities:

— Provide accurate, age-appropriate education

— Address concerns about vaccine safety and effectiveness using current authoritative guidance.

— Check the current Philippine DOH immunization schedule and program eligibility.

— Encourage completion of the recommended vaccine series.

— Reinforce that vaccination is a cancer-prevention strategy.

HPV VACCINE PROPOSED SCHEDULE

Situation	Schedule
Philippine school-based program	Targets eligible grade 4 female learners; catch up may be offered to eligible girls who missed vaccination.
First dose received; second dose due	Give the second dose according to the national program's required interval; the school based guidance specifies at least 6 months after dose 1.
Private or Individual vaccination, age 9-14	Commonly 2 doses, with dose 2 given 6-12 months after dose 1.
Starting at age 15 or older	Commonly 3 doses, often at 0, 1-2, 6 months depending on the vaccine and applicable guidelines.
Immunocompromised individuals	Generally require 3 doses, regardless of age

C. ANXIETY AND DEPRESSION IN ADOLESCENTS

— Adolescent mental health must be treated as physical health.

— Anxiety and depression may affect:

- Sleep
- School attendance
- Relationships
- Concentration
- Nutrition
- Chronic disease management

— WHO identifies anxiety and depression as important causes of illness and disability in adolescence.

Possible Signs of Depression

— Persistent sadness, irritability or loss of interest

— Social withdrawal.

— Changes in sleep or appetite.

— Declining school performance.

— Fatigue, hopelessness or feelings of worthlessness.

— Self-harm or suicidal thoughts.

Nursing Responsibilities

1. Provide a calm, private and nonjudgmental environment.
2. Ask directly about safety, self-harm and suicidal thoughts when concerns arise.
3. Assess urgency, including intent, plan, access to means and recent harm, according to professional training and protocol
4. If there is immediate risk, do not leave the adolescent alone; secure the environment and activate urgent mental health or emergency support.
5. Involve a safe caregiver or support person as appropriate while respecting confidentiality and applicable safeguarding obligations.
6. Arrange appropriate mental health evaluation and follow-up.

Asking directly about suicide does not create suicidal thoughts; it can help identify distress and facilitate support. Immediate danger requires emergency help and continuous supervision until a safe handover.

Never dismiss suicidal statements as attention-seeking. Treat them as clinically significant and assess safety.

HIGH-YIELD COMPARISON ON PEDIATRICS.

Condition	Distinguishing clue	Priority nursing response
Neonatal sepsis	Poor feeding, lethargy, temperature instability	Urgent assessment, stabilization, escalation and prescribed treatment.
Hyperbilirubinemia	Jaundice; timing and bilirubin risk matter	Prompt evaluation and guideline-based treatment.
TOF	Sudden hypercyanotic spell	Urgent help; calm child; knee-chest if appropriate; prescribed emergency care.
Hirschsprung disease / obstruction	Delayed meconium, distension, bilious vomiting	Urgently notify team; monitor obstruction/enterocolitis; follow NPO orders.
Hydrocephalus	Rapid head growth, bulging fontanelle, vomiting	Neurologic assessment and prompt evaluation; monitor shunt.
Hemophilia	Deep bleeding, hemarthrosis, head injury	Control bleeding; assess internal/intracranial bleeding urgently.
Wilms tumor	Painless abdominal mass	Avoid repeated palpation; monitor BP and urine findings.
Severe asthma	Poor air entry; silent chest with distress	Emergency severity assessment and respiratory support/treatment.
DKA	Dehydration, vomiting, Kussmaul breathing	Protocol-based fluids/insulin; monitor potassium and neurologic status.
Severe dengue	Warning signs, cold extremities/poor perfusion	Urgent escalation and carefully reassessed fluid management.
Adolescent suicide risk	Suicidal statement, intent or plan	Immediate safety, continuous supervision and urgent support.

MIDTERMS: CMPA411 (COMPETENCY APPRAISAL 1) - MODULE #10: COMMUNITY HEALTH NURSING

COMMUNITY HEALTH NURSING

— Learned practice discipline with ultimate goal = OPTIMUM LEVEL OF FUNCTIONING

- Health Promotion
- Disease Prevention

Factors Affecting the Health of the Community

1. Characteristics of the People/ Population
2. Location of the Community
3. Social systems within the Community

HEALTH BELIEF MODEL

➢ **Perceived Susceptibilty**

— One's belief regarding the chance of getting a given condition.

➢ **Perceived Severity**

— One's belief regarding the seriousness of a given condition

➢ **Perceived Benefits**

— One's belief in the ability of an advised action to reduce the health risk or seriousness of a given condition.

➢ **Perceived Barriers**

— One's belief regarding the tangible and psychological costs of an advised action.

➢ **Cues to Action**

— Strategies or conditions in one's environment that activate readiness to take action.

➢ **Self-Efficacy**

— One's confidence in one's ability to take action to reduce health risks.

CARE OF INDIVIDUALS, FAMILIES, POPULATION GROUPS AND COMMUNITY

A. NON-COMMUNICABLE DISEASES

— Long-lasting health conditions that do not spread from person to person.

— WHO: More than three-quarters of all NCD deaths, and 86% of the 17 million people who died prematurely, or before reaching 70 years of age, occur in low- and middle-income countries.

Risk Factors:

— Smoking
— Drinking Alcohol
— Unhealthy Diet
— Sedentary Lifestyle

Top Non-Communicable Diseases

— **Ischemic Heart Disease:** The leading cause of death in the country, involving blocked heart arteries.

— **Cancers:** Common types include breast, lung, colorectal, and liver cancer.

— **Strokes (Cerebrovascular Disease):** A major cause of long-term disability and death from interrupted brain blood flow.

— **Diabetes Mellitus:** Primarily Type 2 diabetes, driven by poor diet, high sugar intake, and rising obesity.

— **Chronic Kidney Disease:** A rapidly growing crisis linked heavily to uncontrolled diabetes and hypertension.

— **Hypertension (High Blood Pressure):** Affects over a third of Filipino adults and acts as a major trigger for heart attacks and strokes.

— **Chronic Obstructive Pulmonary Disease (COPD):** A major chronic lung condition tied heavily to smoking and air pollution.

Healthcare Systems and Control

— **Early Screening:** Providing routine checks for high blood pressure, high blood sugar, and early-stage cancers at local clinics.

— **Affordable Treatment:** Ensuring essential medicines for heart attacks, strokes, and diabetes are available via primary health care.

— **Multisectoral Action:** Collaborating across government sectors—including education, agriculture, and urban planning—to create healthier living environments.

— Raise priority accorded to NCDs through international cooperation and advocacy.

— Strengthen national capacity, governance, multisectoral action, and partnerships.

— Reduce the major modifiable risk factors, such as tobacco use, harmful use of alcohol, unhealthy diets, and physical inactivity.

— Develop and implement effective legal frameworks.

— Orient health systems through people-centred health care and universal health coverage.

— Promote high-quality research and development.

— Monitor trends, determinants and progress to achieve global, regional and national targets through evidence-based interventions.

INTEGRATED MANAGEMENT OF CHILDHOOD ILLNESSES (IMCI)

1. ASSESSMENT

— Assess the child by checking first for danger signs (or possible bacterial infection in a young infant), asking questions about common conditions, examining the child, and checking nutrition and immunization status. Assessment includes checking the child for other health problems.

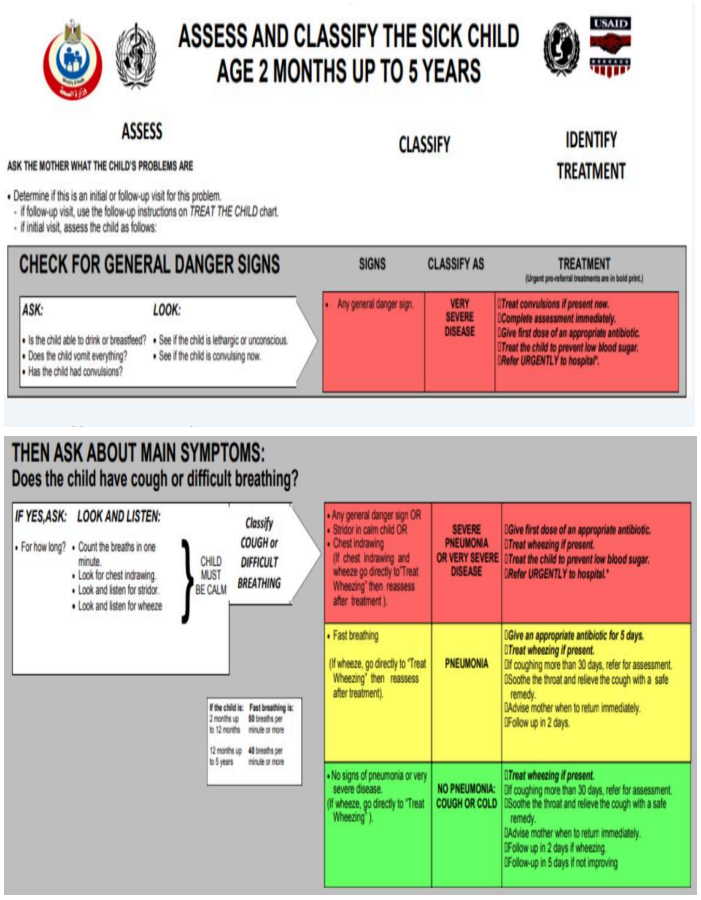
2. CLASSIFICATION

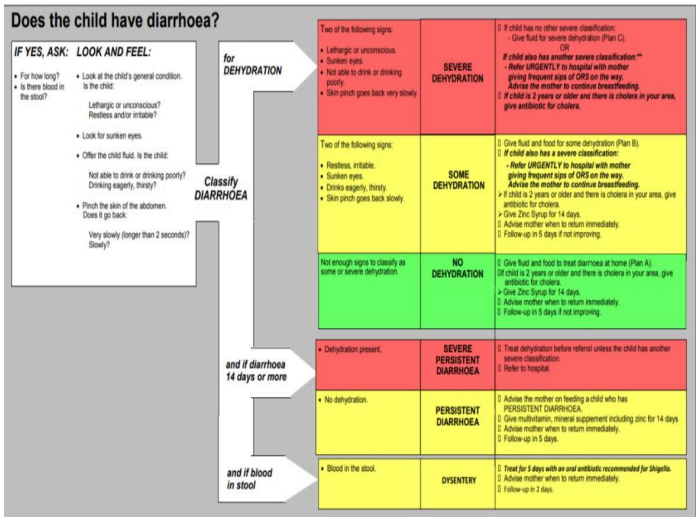
— Classify a child's illnesses using a color-coded triage system. Because many children have more than one condition, each illness is classified according to whether it requires:

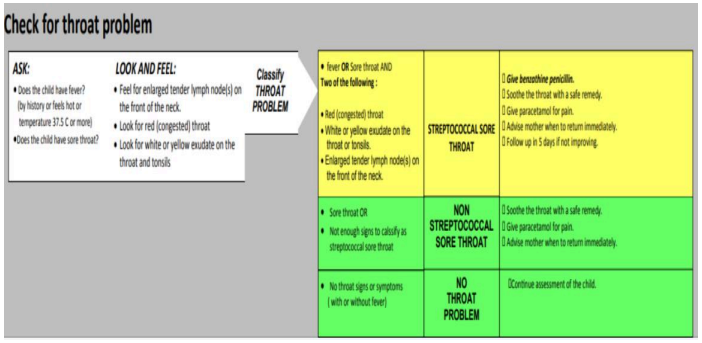
a. urgent pre-referral treatment and referral (RED)
b. specific medical treatment and advice (YELLOW)
c. simple advice on home management (GREEN)

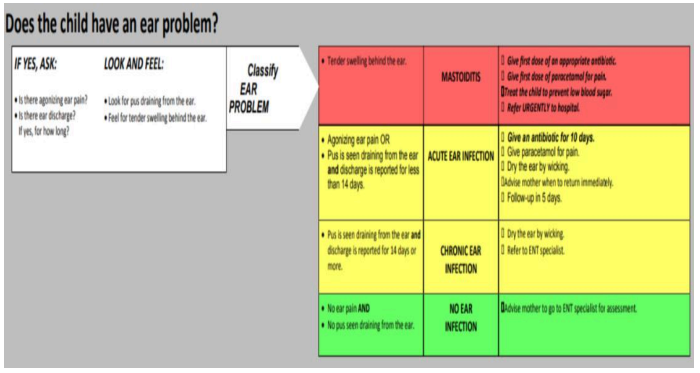
COLOR CODED SYSTEM

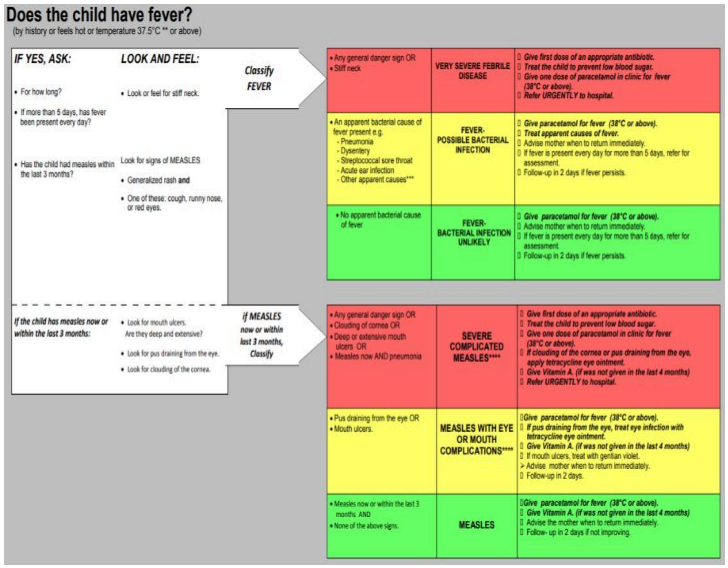
Color Presentation	Classification of Disease	Level of Management
Green	Mild	Home Care
Yellow	Moderate	Management at the RHU
Red	Severe	Referral to hospital

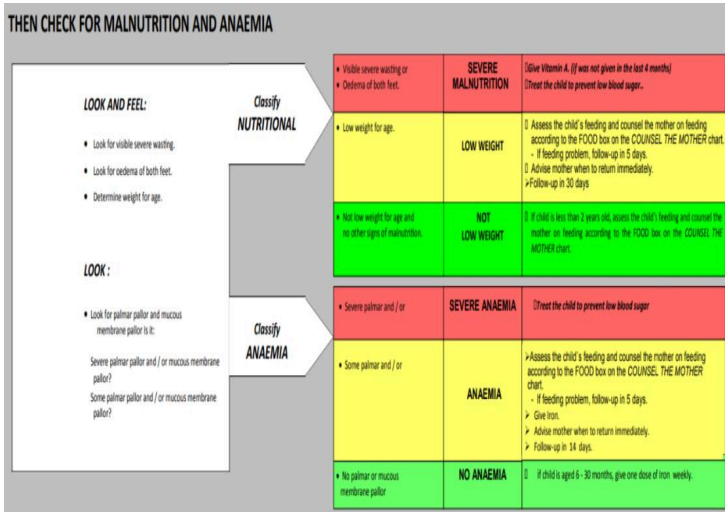


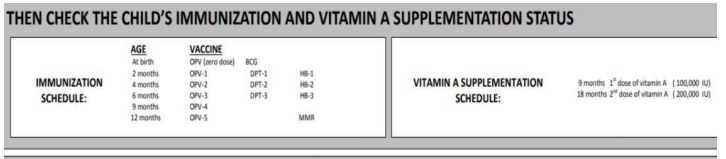


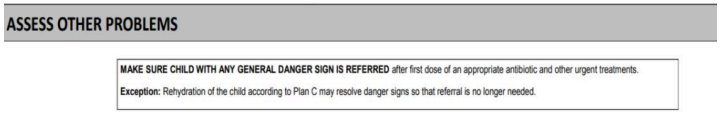












B. COMMUNICABLE DISEASES

— Illnesses that spread from one person, animal, or surface to another.

How They Spread:

1. **Direct contact:** Touching an infected person or exchanging bodily fluids.
2. **Airborne droplets:** Breathing in germs when someone coughs or sneezes.
3. **Contaminated items:** Touching dirty surfaces, food, or water.
4. **Bites:** Getting bit by infected bugs or animals.

Prevention and Care:

1. Wash hands often with soap and water.
2. Get vaccinated to stop severe infections.
3. Take medicines like antibiotics for bacteria or antiviral drugs for viruses.

Communicable Diseases:

➢ **ACUTE RESPIRATORY INFECTION (ARI)**

1. Acute Respiratory Infection (ARI)

— *Causative Agent:* Viruses (Rhinovirus, Respiratory Syncytial Virus [RSV], Adenovirus) or bacteria (Streptococcus pneumoniae, Haemophilus influenzae).

— *Signs & Symptoms:* Cough, fever, sore throat, nasal congestion, tachypnea, and chest indrawing (severe cases).

— *Prevention:* Frequent handwashing, respiratory etiquette, avoiding crowded spaces, and vaccination (Pneumococcal, Influenza).

— *Management:* Supportive care (hydration, rest, antipyretics like paracetamol); targeted antibiotics for confirmed bacterial infections.

2. Influenza (Seasonal)

— *Causative Agent:* Influenza virus types A and B (Orthomyxoviridae family).

— *Signs & Symptoms:* Abrupt onset of high fever, chills, myalgia, severe headache, dry cough, sore throat, and prostration.

— *Prevention:* Annual seasonal influenza vaccination, hand hygiene, wearing masks during outbreaks.

— *Management:* Symptomatic treatment (rest, fluids, NSAIDs/paracetamol); neuraminidase inhibitors (Oseltamivir, Zanamivir) for high-risk patients.

3. Influenza A (H1N1)

— *Causative Agent:* Influenza A virus subtype H1N1 (swine-origin strain).

— *Signs & Symptoms:* Fever, cough, sore throat, body aches, fatigue, with a higher frequency of vomiting and diarrhea than seasonal flu.

— *Prevention:* Annual H1N1/quadrivalent flu vaccine, proper hand sanitation, isolating active cases.

— *Management:* Early administration of oral Oseltamivir (within 48 hours of onset); hydration and antipyretics.

4. Bird Flu (Avian Influenza)

— *Causative Agent:* Avian Influenza A virus strains (primarily H5N1, H7N9).

— *Signs & Symptoms:* High fever, severe respiratory distress, cough, rapid progression to viral pneumonia and Acute Respiratory Distress Syndrome (ARDS); conjunctivitis in some strains.

— *Prevention:* Avoiding contact with sick/dead poultry, thorough cooking of poultry products, PPE for livestock handlers.

— *Management:* Prompt treatment with Oseltamivir or Zanamivir; intensive supportive care including mechanical ventilation in ICU settings.

5. Severe Acute Respiratory Syndrome (SARS)

— *Causative Agent:* SARS Coronavirus (SARS-CoV).

Signs & Symptoms: High fever (>38°C), chills, headache, body aches, followed by dry cough and dyspnea leading to severe hypoxemia.

— *Prevention:* Strict infection control in healthcare settings (N95 respirators, negative pressure rooms), quarantine, contact tracing.

— *Management:* Supportive care, supplemental oxygen, mechanical ventilation for respiratory failure; systemic corticosteroids in severe inflammatory phases.

6. Tuberculosis (TB)

— *Causative Agent:* Mycobacterium tuberculosis (acid-fast bacillus).

— *Signs & Symptoms:* Cough lasting longer than 2–3 weeks, hemoptysis, low-grade evening fever, night sweats, weight loss, and anorexia.

— *Prevention:* BCG vaccination at birth, infection control (ventilation, masks), preventive therapy for Latent TB Infection (LTBI).

— *Management:* First-line Directly Observed Therapy Short-course (DOTS) for 6 months: 2 months of Rifampicin, Isoniazid, Pyrazinamide, Ethambutol (RHZE), followed by 4 months of RH.

7. Pertussis (Whooping Cough)

— *Causative Agent:* Bordetella pertussis (Gram-negative bacterium).

— *Signs & Symptoms:* Starts as mild cold symptoms, progressing to severe paroxysmal coughing fits ending in a high-pitched "whoop" and post-tussive vomiting.

— *Prevention:* DTaP/Tdap vaccination (including maternal immunization during the third trimester); post-exposure prophylaxis for contacts.

— *Management:* Macrolide antibiotics (Azithromycin or Erythromycin) to reduce infectivity; supportive airway and hydration management.

8. Diphtheria

— *Causative Agent:* Corynebacterium diphtheriae (toxin-producing bacterium).

— *Signs & Symptoms:* Thick grey pseudomembrane covering tonsils/pharynx, sore throat, cervical lymphadenopathy ("bull neck"), fever, and airway compromise.

— *Prevention:* DTaP/Pentavalent/Tdap vaccination; booster doses every 10 years.

— *Management:* Immediate administration of Diphtheria Antitoxin (DAT) and antibiotics (Penicillin G or Erythromycin); airway maintenance.

➢ GASTROINTESTINAL AND WATERBORNE INFECTIONS

9. Cholera

— *Causative Agent:* Vibrio cholerae (toxigenic bacterium).
— *Signs & Symptoms:* Rapid onset of profuse, painless "rice-water" diarrhea, severe vomiting, rapid dehydration, and severe muscle cramps.
— *Prevention:* Clean water supply (boiling/chlorination), proper sanitation, food hygiene, Oral Cholera Vaccine (OCV).
— *Management:* Immediate rehydration with Oral Rehydration Salts (ORS) or intravenous fluids (Ringer's Lactate); antibiotics (Doxycycline or Azithromycin) in severe cases.

10. Diarrhea (Infectious Gastroenteritis)

— *Causative Agent:* Rotavirus, Norovirus, Escherichia coli, Salmonella, Shigella, or Giardia lamblia.
— *Signs & Symptoms:* Frequent loose or watery stools, abdominal cramping, nausea, vomiting, fever, and dehydration signs.
— *Prevention:* Rotavirus vaccine, thorough handwashing with soap, safe food handling, safe drinking water.
— *Management:* Fluid and electrolyte replacement via ORS; Zinc supplementation in young children; targeted antimicrobials for identified bacterial/parasitic pathogens.

11. Typhoid Fever

— *Causative Agent:* Salmonella enterica serovar Typhi.
— *Signs & Symptoms:* Sustained step-ladder fever, severe headache, abdominal pain, constipation or diarrhea, hepatosplenomegaly, and "rose spots" rash on trunk.
— *Prevention:* Injectable Vi capsular or oral Ty21a vaccines, strict food and water precautions, proper hand hygiene.
— *Management:* Antibiotic therapy (Ceftriaxone, Azithromycin, or Ciprofloxacin depending on resistance patterns); adequate hydration.

➢ VECTOR-BORNE AND ZOONOTIC DISEASE

12. Dengue

— *Causative Agent:* Dengue Virus (DENV-1, 2, 3, 4; Flaviviridae).
— *Signs & Symptoms:* High fever, severe retro-orbital pain, myalgia/arthralgia ("breakbone fever"), petechiae; Severe Dengue presents with fluid accumulation, plasma leakage, mucosal bleeding, and severe organ impairment.
— *Prevention:* Mosquito vector control (eliminating standing water, larvicides), repellents, bed nets, wearing protective clothing.
— *Management:* Strict fluid management, rest, monitoring hematocrit/platelets, paracetamol for fever. Avoid NSAIDs/Aspirin due to bleeding risk.

13. Malaria

— *Causative Agent:* Plasmodium parasites (P. falciparum, P. vivax, P. ovale, P. malariae, P. knowlesi) transmitted by female Anopheles mosquitoes.
— *Signs & Symptoms:* Cyclic paroxysms of shaking chills, high fever, profuse sweating, headache, anemia, and splenomegaly.
— *Prevention:* Insecticide-treated bed nets (ITNs), indoor residual spraying (IRS), chemoprophylaxis for travelers, RTS, S / R21 vaccines.
— *Management:* Artemisinin-based Combination Therapies (ACTs) for P. falciparum; Chloroquine plus Primaquine for P. vivax to eradicate liver hypnozoites.

14. Leptospirosis

— *Causative Agent:* Leptospira interrogans (spirochete bacterium).
— *Signs & Symptoms:* High fever, intense calf and lumbar muscle pain, conjunctival suffusion (red eyes without exudate); Severe form (Weil's disease) causes jaundice, renal failure, and bleeding.
— *Prevention:* Avoiding wading in floodwaters or soil contaminated with animal urine, wearing protective footwear/gloves, rodent control.
— *Management:* Early treatment with Doxycycline (mild cases) or IV Penicillin G / Ceftriaxone (severe cases); hemodialysis for renal failure.

15. Rabies

— *Causative Agent:* Rabies virus (Lyssavirus genus).
— *Signs & Symptoms:* Early paresthesia/pain at bite site; Encephalitic: hydrophobia, aerophobia, agitation; Paralytic: ascending paralysis leading to coma and death.
— *Prevention:* Animal/pet vaccination, avoiding stray animals; immediate Post-Exposure Prophylaxis (PEP: rabies vaccine + Rabies Immunoglobulin [RIG]) post-exposure.
— *Management:* Virtually 100% fatal once clinical symptoms develop; management is supportive/palliative.

16. Ebola Virus Disease (EVD)

— *Causative Agent:* Ebola virus (Filoviridae family).
— *Signs & Symptoms:* Sudden fever, intense weakness, myalgia, sore throat, followed by vomiting, diarrhea, maculopapular rash, impaired liver/kidney function, and internal/external hemorrhaging.
— *Prevention:* Ebola vaccination (Ervebo), strict Personal Protective Equipment (PPE) for healthcare workers, safe burial protocols.
— *Management:* Specific monoclonal antibody treatments (Inmazeb, Ebanga); aggressive rehydration, electrolyte restoration, and organ support.

➢ EXANTHEMATOUS, ENTERIC & SYNTHETIC INFECTIOUS DISEASE

17. Chickenpox

— *Causative Agent:* Varicella-Zoster Virus (VZV).
— *Signs & Symptoms:* Pruritic rash evolving in crops from macules to papules, itchy fluid-filled vesicles, and crusts; accompanied by low-grade fever and malaise.
— *Prevention:* Varicella vaccine (2-dose series); isolation of infected individuals until all lesions have crusted over.
— *Management:* Symptomatic care (calamine lotion, antihistamines, paracetamol); oral Acyclovir for high-risk individuals. Avoid aspirin due to Reye syndrome risk.

18. Hand, Foot, and Mouth Disease (HFMD)

— *Causative Agent:* Coxsackievirus A16 or Enterovirus 71 (EV-A71).
— *Signs & Symptoms:* Fever, painful ulcerative oral lesions (enanthem), maculopapular rash or blisters on palms, soles, and buttocks.
— *Prevention:* Proper handwashing, disinfecting daycare/household surfaces, avoiding contact with blisters or oral secretions.
— *Management:* Supportive care with hydration, soft cool foods, and analgesics/antipyretics (Paracetamol, Ibuprofen).

19. Measles (Rubeola)

— *Causative Agent:* Measles virus (Morbillivirus genus).
— *Signs & Symptoms:* High fever, the "3 Cs" (Cough, Coryza, Conjunctivitis), Koplik spots on buccal mucosa, followed by an erythematous maculopapular rash spreading face-downward.
— *Prevention:* MMR vaccine (2 doses); isolation during the infectious period.
— *Management:* Supportive care, fluid replacement, antipyretics; High-dose Vitamin A administration on consecutive days to prevent complications (e.g., blindness, pneumonia).

20. Meningococcemia

— *Causative Agent:* Neisseria meningitidis (Gram-negative diplococcus).
— *Signs & Symptoms:* Sudden high fever, severe headache, petechial or purpuric skin rash, stiff neck, altered mental status, and rapid septic shock (Waterhouse-Friderichsen syndrome).
— *Prevention:* Meningococcal conjugate vaccines (MenACWY, MenB); chemoprophylaxis (Rifampin, Ciprofloxacin, or Ceftriaxone) for close contacts.
— *Management:* Emergency administration of IV antibiotics (Ceftriaxone or Cefotaxime); aggressive resuscitation, vasopressors, and ICU care.

21. Poliomyelitis

— *Causative Agent:* Poliovirus (Types 1, 2, and 3; Enterovirus genus).
— *Signs & Symptoms:* Most infections are asymptomatic; Non-paralytic: fever, sore throat, neck stiffness; Paralytic: acute flaccid paralysis (asymmetrical), muscle atrophy, respiratory arrest.
— *Prevention:* Inactivated Polio Vaccine (IPV) and Oral Polio Vaccine (OPV).
— *Management:* No specific antiviral treatment available; supportive management (physical therapy, pain relief, mechanical ventilation for respiratory paralysis).

22. Leprosy (Hansen's Disease)

— *Causative Agent:* Mycobacterium leprae or Mycobacterium lepromatosis.
— *Signs & Symptoms:* Hypopigmented or reddish skin lesions with definitive loss of sensation, peripheral nerve enlargement, muscle weakness, and peripheral anesthesia.
— *Prevention:* Early detection and treatment of index cases to stop transmission; BCG vaccination provides partial protection.
— *Management:* Standard Multi-Drug Therapy (MDT) combining Rifampicin, Dapsone, and Clofazimine; nerve function monitoring and physical rehabilitation.

23. HIV/AIDS

— *Causative Agent:* Human Immunodeficiency Virus (HIV-1 and HIV-2).
— *Signs & Symptoms:* Acute: Flu-like seroconversion illness; Chronic/AIDS: Progressive decline in CD4 T-cell count leading to opportunistic infections (e.g., Pneumocystis pneumonia, TB, Kaposi sarcoma), chronic diarrhea, severe weight loss.
— *Prevention:* Barrier contraceptives (condoms), Pre-Exposure Prophylaxis (PrEP), Post-Exposure Prophylaxis (PEP), sterile syringe programs, and Prevention of Mother-to-Child Transmission (PMTCT).
— *Management:* Lifelong Antiretroviral Therapy (ART) using combination regimens to suppress viral load; prophylaxis/treatment for opportunistic infections.

COMMUNICABLE DISEASE CONTROL

— Communicable disease control, elimination, and eradication form a step-by-step public health framework. Control lowers disease numbers to a safe level, elimination stops transmission in a specific area, and eradication permanently ends global transmission worldwide.

Disease Control

— ***Goal:*** Lower new cases, sickness, and death to a low, acceptable level.
— ***Action:*** Needs ongoing public health work, like treatment and safe water, to keep numbers down.
— ***Example:*** Managing seasonal flu or ongoing World Health Organization tuberculosis programs.

Disease Elimination
— ***Goal:*** Reduce new infections to zero in a specific region, country, or continent.
— ***Action:*** Requires strong local tracking and fast responses to stop local spread.
— ***Example:*** Stopping local polio or neonatal tetanus in specific countries.
Disease Eradication
— ***Goal:*** Permanently wipe out the infectious agent so no one gets sick anywhere on Earth.
— ***Action:*** Stops all medical interventions because the germ no longer exists in nature.
— ***Example:*** Smallpox is the only human disease fully wiped out this way.

Key Goals of Family Care
— Care of normal families in a community setting focuses on health promotion and disease prevention. Community health nurses use a systematic nursing process—assessment, diagnosis, planning, implementation, and evaluation—to support family wellness, manage developmental stages, and encourage healthy lifestyles through local health centers, schools, and home visits.
— **Promote wellness:** Build healthy habits like good nutrition and regular exercise.
— **Prevent illness:** Provide immunizations and routine health screenings.
— **Support transitions:** Guide families through normal life changes like birth, growth, and aging.
— **Strengthen coping:** Help members solve daily stress and minor conflicts.

Common Nursing Strategies
— **Home visits:** Face-to-face meetings in the house to check living conditions.
— **Health education:** Teach disease prevention and hygiene.
— **Clinic services:** Offer check-ups, vaccines, and family planning at local centers.
— **Referral linkage:** Connect families to social welfare and external medical resources.

What are the responsibilities of a family in the community?
— **Economic support:** providing basic needs, such as food, shelter, clothing, etc.
— **Emotional support:** providing love, comfort, intimacy, companionship, nurturing, belongingness, etc.
— **Socialization of children:** rearing children, parenting, helping children function to the best of their abilities within their society.

Identifying Risk Factors
— Caring for families at risk in a community setting involves a family-centered nursing and public health approach that identifies environmental, economic, and behavioral hazards. Nurses and local health workers use tools like home visits and risk assessments to lower stress, build coping skills, and connect households to vital social programs.
— **Poverty and Food Insecurity:** Lack of steady jobs, low income, and poor nutrition.
— **Neighborhood Stress:** High crime rates, unsafe housing, and drug or alcohol abuse.
— **Life Transitions:** Major events like divorce, death, chronic illness, or birth.
— **Family Dynamics:** Poor communication, lack of parental supervision, or history of abuse

Core Community Interventions
— **Home Visits:** Face-to-face professional contact in the home to evaluate living conditions and give direct care.
— **Family Assessment:** Using tools like genograms and ecomaps to chart health history and support networks.
— **Resource Linking:** Connecting parents to assistance programs like food stamps, WIC, or local health clinics.
— **Parenting Education:** Teaching positive discipline, child development, and peaceful conflict resolution.

PRIORITIZATION

Criteria	Highest Score	Weight
Name of the Problem: — Wellness State - 3 — Health Deficit - 3 — Health Threat - 2 — Foreseeable Crisis - 1	3	1
Modifiable of the Problem: — Easily modifiable - 2 — Partially modifiable - 1 — Not modifiable - 0	2	2
Preventive Potential: — High - 3 — Moderate - 2 — Low - 1	3	1
Salience Scale: — Condition or problem, needing immediate attention - 2 — A condition or problem not needing immediate attention - 1 — Not perceived as a problem or condition needing change - 0	2	1

Formula: Scored according to criteria/ highest score "weight = actual score.

Priority Setting and Assessment
— Caring for population groups with other health priorities means identifying secondary or specific health needs, analyzing community barriers, and planning targeted interventions when major or emergency issues take initial focus. It relies on priority-setting tools, active group assessments, and resource allocation.

— **Data gathering:** Use surveys, interviews, and observations to find hidden or overlooked health gaps in a group.
— **Scoring methods:** Apply the Hanlon Method or a Prioritization Matrix to rank issues by how severe and fixable they are.
— **Core criteria:** Check problem significance, community awareness, and the cost of reducing risks.

Planning and Interventions
— **Barrier removal:** Find out why a group misses standard care, such as lack of local transport or high costs.
— **Targeted action:** Design specific health teachings or mobile screenings that fit the group's daily life.
— **Evaluation:** Measure if the short-term goals improved health results and reduced group disparities.

SUSTAINABLE DEVELOPMENT GOALS



REPUBLIC ACT NO. 11223

— An Act Instituting Universal Health Care for All Filipinos, Prescribing Reforms in the Health Care System, and Appropriating Funds Therefor
— Section 1. This Act shall be known as the "Universal Health Care Act".
— Section 2. Declaration of Principles and Policies. - It is the policy of the State to protect and promote the right to health of all Filipinos and instill health consciousness among them.
— Section 5. Population Coverage. - Every Filipino citizen shall be automatically included into the NHIP, hereinafter referred to as the Program. Essential health benefit package refers to a set of individual-based entitlements covered by the National Health Insurance Program (NHIP) which includes primary care; medicines, diagnostics and laboratory; and preventive, curative, and rehabilitative services;

Core Features of the UHC Law
— **Automatic Inclusion:** Every Filipino is an automatic member of PhilHealth, categorized as either a direct or indirect contributor.
— **Expanded Services:** Coverage includes preventive, promotive, diagnostic, curative, rehabilitative, and palliative care.
— **Primary Care Focus:** Emphasizes outpatient benefits, free consultations, and basic laboratory tests to stop illnesses early.
— **System Integration:** Combines local and national health systems into coordinated provincial and city-wide networks.

1. COMMUNITY ASSESSMENT (The "Data" Phase)
— Students often struggle to see the community as a single client. Educators must shift their focus from individual bedside charts to population-level data.

Key Concepts to Teach:
1. **The Community Core:** People, history, demographics, values, and beliefs.
2. **Subsystems:** Physical environment, health/social services, economy, transportation, safety, politics, communication, and recreation.
3. **Data Types:** Quantitative (census data, morbidity rates) vs. Qualitative (key informant interviews).

2. COMMUNITY DIAGNOSIS (The "Analysis" Phase)
— Students must learn that a community diagnosis is not a medical diagnosis. It focuses on aggregates, risk factors, and environmental etiologies.

Key Concepts to Teach:
— NANDA/Omaha System for CHN: Structuring diagnoses at the community level.
— Risk of/Increased incidence of... (Health problem)
— Among... (Specific aggregate/population)
— Related to... (Etiology/contributing factors)
— As evidenced by... (Assessment data/statistics)

THE THREE-PART CHN DIAGNOSIS FORMULA

Public health nursing commonly uses a modified version of the NANDA/Omaha system pattern. This structure forces students to link the problem directly to its environmental root cause and supporting data.
— Part 1: **The Problem / Risk Statement:** The specific health deficit, threat, or wellness state.
— Part 2: **The Target Aggregate:** The specific group, demographic, or geographic boundary affected.
— Part 3: **Related To (Etiology):** The socio-economic, environmental, behavioral, or political root causes.
— Part 4: **As Evidenced By (Signs/Symptoms):** The concrete qualitative or quantitative data gathered during the assessment phase.

Classification of Community Problems
Before writing a diagnosis, students should categorize their findings into one of three typologies:
— **Health Deficits:** Instances of disease, disability, or death within the community (e.g., high scabies rates).
— **Health Threats:** Conditions that promote disease or injury but haven't caused a major outbreak yet (e.g., open drainage systems, low vaccination rates)
— **Foreseeable Crisis / Stress Points:** Anticipated situations that can demand community adjustment (e.g.,

upcoming typhoon season, sudden closure of the area's main livelihood factory).

A. PLANNING (THE "DESIGN" PHASE)

— The focus here is moving students away from "individual patient care plans" toward "sustainable population-level programs."

Key Concepts to Teach:

— **Community Mobilization:** Involving community leaders (e.g., local health workers, barangay officials) so the program is done with the community, not just to them.

— **Levels of Prevention:** Primary (health promotion/vaccines), Secondary (screenings), and Tertiary (rehabilitation/chronic disease management).

— **SMART Objectives:** Ensuring goals are Specific, Measurable, Achievable, Realistic, and Time-bound at the population level.

B. INTERVENTION (THE "IMPORTANCE" PHASE)

— This is the action phase where students practice public health nursing roles: educator, advocate, coordinator, and clinician.

Key Concepts to Teach:

— **Health Literacy:** Adapting health education materials to the reading and cultural level of the target population.

— **Sustainability:** Designing interventions that the community can maintain after the students' clinical rotation ends.

C. EVALUATION (THE "APPRAISAL" PHASE)

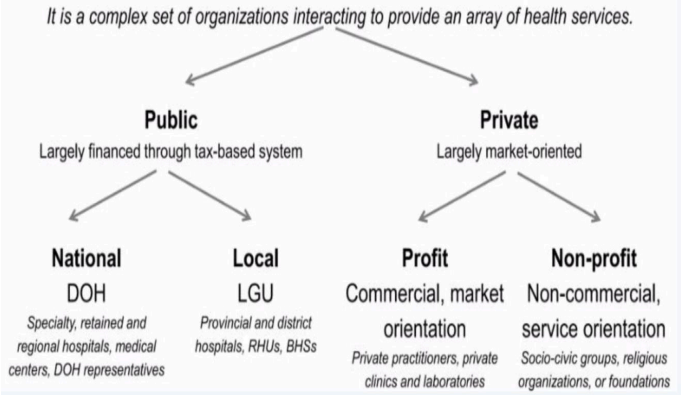
— Students frequently confuse evaluating their own performance with evaluating the program's impact. Faculty must clarify this distinction.

Key Concepts to Teach:

— **Formative Evaluation:** Assessing the process during implementation (e.g., "Did we have enough brochures? Did people show up?").

— **Summative Evaluation:** Assessing the final impact after the program (e.g., "Did the community's knowledge scores increase? Did incidence rates drop?").

Philippine Health Care Delivery System



PHASES OF EPIDEMIOLOGICAL APPROACH

1. DESCRIPTIVE EPIDEMIOLOGY

— Describes frequency and distribution of disease in a given population.

— Observes and records existing patterns of occurrence of disease.

— Describes disease as: person, place and time characteristics.

2. ANALYTICAL EPIDEMIOLOGY

— Attempts to analyze the causes or determinants of disease through hypothesis testing.

3. INTERVENTION OR EXPERIMENTAL EPIDEMIOLOGY

— Answers questions about the effectiveness of new methods for controlling diseases or for improving underlying conditions.

4. EVALUATION EPIDEMIOLOGY

— Attempts to measure the effectiveness of different health services and programs.

INCIDENCE VS. PREVALENCE

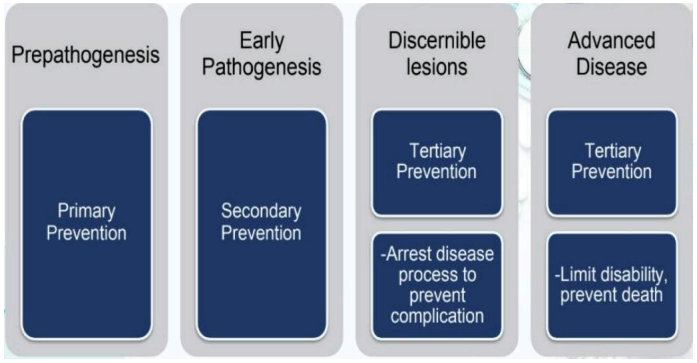
$$\text{Incidence Rate} = \frac{\text{Number of NEW cases during a specific period}}{\text{Population at risk during that period}} \times k$$

— **Incidence:** Measures the risk of contracting a condition. It looks exclusively at newly diagnosed cases over a given timeframe.

— **Prevalence:** Measures the overall burden of a disease in a population at a specific point or period in time (includes both new and existing cases)

$$\text{Prevalence Rate} = \frac{\text{Total existing cases at a given time}}{\text{Total population at that time}} \times k$$

NATURAL LIFE HISTORY OF DISEASE AND APPLICATION OF LEVELS OF DISEASE PREVENTION



A. PRIMARY LEVEL OF CARE

— **First point of contact:** Serves as the initial entry point into the health system for individuals and families.

— **Facilities:** Includes Barangay Health Stations (BHS), Rural Health Units (RHUs), and city or municipal health offices, as well as private clinics.

— **Services:** Focuses on health promotion, disease prevention, immunizations, maternal and child care, and basic treatment of common illnesses.

B. SECONDARY LEVEL OF CARE

— **Step-up care:** Involves specialized treatment for patients requiring hospitalization or acute intervention.
— **Facilities:** Comprises non-departmentalized or smaller district and provincial hospitals.
— **Services:** Handles common diseases and injuries that need clinical care, general surgery, minor specialist consultations, and emergency inpatient services.

C. TERTIARY LEVEL OF CARE

— **Advanced specialty care:** Delivers high-tech, sophisticated, and comprehensive medical services for severe or rare health conditions.
— **Facilities:** Includes regional medical centers, national specialty centers, and large teaching or training hospitals (such as the Philippine General Hospital).
— **Services:** Provides highly specialized surgical procedures, sub-specialty care, advanced diagnostic testing, rehabilitation, and medical research.

PATTERN OF DISEASES OCCURENCE

➢ EPIDEMIC

— Situation when there is a marked upward fluctuation in disease incidence.

➢ ENDEMIC

— Habitual presence of disease in a given geographic location.

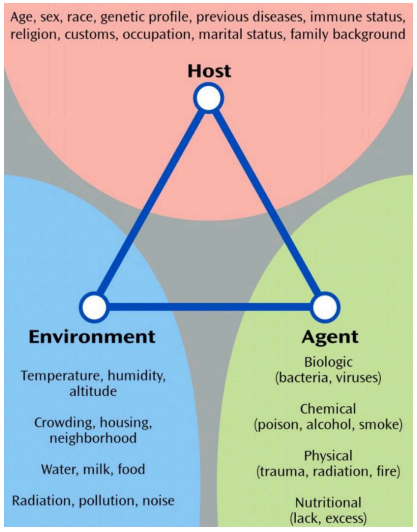
➢ SPORADIC

— When disease occur every now and then affecting only a small number of people relative to the total population.

➢ PANDEMIC

— Worldwide occurence

Element	Definition	Examples
Agent	The cause of the disease or health condition	Biological (viruses, bacteria), Chemical (toxins, drugs), Physical (trauma, radiation)
Host	The living species susceptible to or hosting the agent.	Age, biological sex, genetic profile, immune status, lifestyle, pre-existing conditions.
Environment	External conditions that allow disease transmission	Living conditions, sanitation, climate, social structure, healthcare access.



DEMOGRAPHY

— The statistical study of human populations, including their size, structure, distribution, and how they change over time.

VITAL STATISTICS

Official government records and data tracking crucial human life events and population changes.
— To determine success or failure of specific health problems.
— To promote health legislation at local and national level.
— To develop policies and procedure at state and center.

CRUDE BIRTH RATE

$$\frac{\text{Total of live births registered in a given calendar year}}{\text{Estimated population as of July 1 of same year}} \times 1000$$

GENERAL FERTILITY RATE

$$\frac{\text{\# of registered livebirths in a year}}{\text{Midyear population of women } 15-44 \text{ years of age}} \times 1000$$

MORBIDITY (DISEASE) RATE

$$\frac{\text{No. of cases of residents with illness in an area in a calendar year}}{\text{Population in the area in the same year}} \times 1000$$

INCIDENCE RATE

$$\frac{\text{Number of new health} - \text{related events or cases of disease}}{\text{Number of people exposed to risk during this period}}$$

ATTACK RATE

$$\frac{\text{The cumulative incidence of infection in a group observed during an epidemic}}{\text{Number of people exposed}} \times 100$$

PREVALENCE RATE

$$\frac{\text{Total number of all individuals who have an attribute or disease at a time}}{\text{Population at risk of having the attribute or disease at this point or period of time}}$$

PHILIPPINE HEALTH CARE LAWS

— **Republic Act 3573** - Law on reporting of Communicable Disease
— **Republic Act 4226** - Hospital Licensure Act requires all hospitals to be licensed before it can operate.
— **Republic Act 6425** - Dangerous Drug Act of 1992
— **Republic Act 6675** - Generics Act of 1988
— **Republic Act 7600** - Rooming In and Breastfeeding Act of 1992
— **Republic Act 7875** - National Health Insurance Act of 1995
— **Republic Act 9288** - Newborn Screening Act
— **Presidential Decree 996** - Provides for compulsory basic immunization for children and infants below 8 years of age.
— **Executive Order 51** - The Milk Code

COMMUNITY ORGANIZING PARTICIPATORY ACTION RESEARCH (COPAR)

— A social development approach that aims to transform the apathetic, individualistic and voiceless poor into dynamic, participatory and politically responsive community.
— A collective, participatory, transformative, liberative, sustained and systematic process of building people's organizations by mobilizing and enhancing the capabilities and resources of the people for the resolution of their issues and concerns towards effecting change in their existing oppressive and exploitative conditions (1994 National Rural Conference)
— A process by which a community identifies its needs and objectives, develops confidence to take action in respect to them and in doing so, extends and develops cooperative and collaborative attitudes and practices in the community (Ross 1967).
— A continuous and sustained process of educating the people to understand and develop their critical awareness of their existing condition, working with the people collectively and efficiently on their immediate and long-term problems, and mobilizing the people to develop their capability and readiness to respond and take action on their immediate needs towards solving their long-term problems (CO: A manual of experience, PCPD)

Importance of COPAR

1. COPAR is an important tool for community development and people empowerment as this helps the community workers to generate community participation in development activities.
2. COPAR prepares people/clients to eventually take over the management of a development programs in the future.
3. COPAR maximizes community participation and involvement; community resources are mobilized for community services.

Principles of COPAR

1. People, especially the most oppressed, exploited and deprived sectors are open to change, have the capacity to change and are able to bring about change.
2. COPAR should be based on the interest of the poorest sectors of society
3. COPAR should lead to a self-reliant community and society.

COPAR Process

— A progressive cycle of action-reflection action which begins with small, local and concrete issues identified by the people and the evaluation and the reflection of and on the action taken by them.
— Consciousness through experimental learning central to the COPAR process because it places emphasis on learning that emerges from concrete action and which enriches succeeding action.
— COPAR is participatory and mass-based because it is primarily directed towards and biased in favor of the poor, the powerless and oppressed.
— COPAR is group-centered and not leader-oriented. Leaders are identified, emerge and are tested through action rather than appointed or selected by some external force or entity.

COPAR Phases of Process

1. PRE-ENTRY PHASE

— Is the initial phase of the organizing process where the community/organizer looks for communities to serve/help.
— It is considered the simplest phase in terms of actual outputs, activities and strategies and time spent for it

Activities include:

— Designing a plan for community development including all its activities and strategies for care development.
— Designing criteria for the selection of site
— Actually selecting the site for community care

2. ENTRY PHASE

— Sometimes called the social preparation phase as to the activities done here includes the sensitization of the people on the critical events in their life, innovating them to share their dreams and ideas on how to manage their concerns and eventually mobilizing them to take collective action on these.

— This phase signals the actual entry of the community worker/organizer into the community. She must be guided by the following guidelines however.
— Recognizes the role of local authorities by paying them visits to inform them of their presence and activities.
— The appearance, speech, behavior and lifestyle should be in keeping with those of the community residents without disregard of their being role models.
— Avoid raising the consciousness of the community residents; adopt a low-key profile.

3. ORGANIZATION BUILDING PHASE

— Entails the formation of more formal structures and the inclusion of more formal procedures of planning, implementation, and evaluating community-wide activities. It is at this phase where the organized leaders or groups are being given trainings (formal, informal, OJT) to develop their skills and in managing their own concerns/programs.

4. SUSTENANCE AND STRENGTHENING PHASE

— Occurs when the community organization has already been established and the community members are already actively participating in community-wide undertakings. At this point, the different communities setup in the organization building phase are already expected to be functioning by way of planning, implementing and evaluating their own programs with the overall guidance from the community-wide organization.

Strategies used may include:
— Education and training
— Networking and linkaging
— Conduct of mobilization on health and development concerns
— Implementing of livelihood projects
— Developing secondary leaders

VACCINE PROCUREMENT IN THE COMMUNITY

ELIGIBLE POPULATION – the group of people, described demographically, who are qualified to receive program services.
ELIGIBLE POPULATION – total population x 2.7%

VACCINE	FIRST DOSE	DOSAGE	# OF DOSES	INTERVAL/ NEXT DOSE	ROUTE	PROTECTION AGAINST
BCG	At birth	0.05ml	1	None	ID	Tuberculosis
HEPATITIS B	At birth	0.5ml	3	1-2 months then 6-18 months	IM	Hepatitis B
DPT	6 weeks	0.5ml	3	4 weeks	IM	Diphtheria, Pertussis, Tetanus
OPV	6 weeks	2-3 drops	3	4 weeks	PO	Poliomyelitis
HiB	6 weeks or 2 months	0.5 ml	3	4 weeks	IM	Haemophilus Influenza type B infection
PCV	6 weeks	0.5 ml	3	4 weeks	IM	Streptococcus pneumoniae
MMR	9 months	0.5 ml	2	1 year	SC	Measles, Mumps, & Rubella

WASTAGE FACTOR (wf) – the factor (number) that you multiply your estimated vaccine needs by, to allow for some doses being wasted.

Monthly Vial Needs = EP x #of doses x wf/dose per vial/ #of months

VACCINES	3 OF DOSES	WASTAGE FACTOR	3 OF DOSES PER VIAL/AMP
BCG	1	2.5	20
HEPA B	1	1.1	10
PENTAVALENT (DPT, HepaB, HiB)	3	1.1	1
PCV	3	1.1	1
OPV	3	1.67	20
IPV	1	1.25	10
MMR	1	1.1	5